Teaching Notes: Statistical Modelling

Prof. Fulvia Pennoni

Statistical Modelling

University of Milano-Bicocca

fulvia.pennoni@unimib.it

The quiet statisticians have changed our world;

not by discovering new facts or technical developments,

but by changing the ways that we reason,

experiment and form our opinions ....

- Ian Hacking -

April 29, 2026

Contents

1 Introduction **5**

1.1 Statistical inference 7

1.2 Review of some basic definitions 11

2 A review of random variables, distributions and measures of association **16**

2.1 Gaussian distribution 18

2.2 Student-t 20

2.3 Empirical cumulative distribution function 23

2.4 Quantile-Quantile plot 23

2.5 Discrete random variables: Expected value and variability 24

2.6 Joint and conditional distributions 25

2.7 Measures of associations 28

2.8 Partial linear correlation coefficient 31

2.8.1 Scatterplot to visualize bivariate associations 32

2.9 Bivariate and multivariate Gaussian distribution 33

2.9.1 Bivariate Gaussian distribution 33

2.9.2 Multivariate Gaussian distribution 36

2.9.3 Singular value decomposition 38

3 Statistical inference: Estimation **39**

3.1 Review of the properties of estimators 39

3.2 Likelihood function and maximum likelihood estimation 41

3.3 Bayesian methods 46

3.4 Inference based on resampling (bootstrap) method 47

3.4.1 Confidence intervals: Percentile Method 51

3.5 Summary 53

3.6 Jackknife 54

4 Multiple linear regression **55**

4.1 Model specification 56
4.1.1 Matrix notation 58
4.2 Model specification with two explanatory variables 59
4.3 Least squares estimation method 60
4.4 Residuals 63
4.5 Decomposing variability: Analysis of variance 64
4.5.1 Multiple R-Squared 66
4.6 Diagnostics for unusual and influential observations 67

5 Inference for the multiple linear regression model 71
5.1 Properties of the least squares estimators 73
5.2 Inference on the model parameters 75
5.2.1 Testing that all effects are equal to zero: the F-test 76
5.2.2 Remarks on the chi-squared and F distributions 79
5.2.3 Confidence intervals for the regression parameters 81
5.3 t-test for each regression parameter 81
5.4 Multicollinearity: Nearly redundant explanatory variables 83
5.4.1 Variance Inflation Factor 84

6 Variable selection: Data-driven procedures 85

7 Prediction 88
7.1 Bootstrapping 92

8 Strategies for model building 92

9 Categorical explanatory variables 95

10 Extension of the classical linear model 97

11 Recall of Bernoulli and Binomial distributions 98
11.1 Bernoulli distribution 98
11.2 Binomial distribution 98

12 Logistic regression for binary data **101**

12.1 Inference 104

13 Model-based clustering and classification **107**

13.1 Finite mixture models 108

13.2 Mixtures of Gaussian distributions 109

13.3 Parsimonious covariance decomposition 110

13.4 Maximum likelihood inference 114

© *F. Pennoni*

All rights reserved. Students are not allowed to reproduce the following material

Citation: Pennoni F. (2026). Teaching notes: Statistical Modelling. Department of Statistics and Quantitative Methods, University of Milano-Bicocca.

1 Introduction

Statistics is proposed as a set of methods that allow the study of collective phenomena and it can be qualified as the science of *learning from experience* (Efron and Hastie, 2021). This science has almost 250-year history. In 1600 in England, the inductive empirical method began to be applied in the social sciences. In 1741, the first statistical tables appeared in Denmark. Sir Thomas Bayes (1701-1761) proposed the Bayes'rule published later in 1763. Using mathematics not in its abstract content but in relation to concrete problems gave a real impetus to the development of statistical science. Karl Pearson (1857-1936), an English mathematician and biostatistician, was one of the pioneers in this field (he published a paper on the chi-squared distribution) along with Sir Ronald Aylmer Fisher (1890-1962, publishing paper on maximum likelihood estimation). The Russian mathematician Andrey Andreyevich Markov (1856-1922), in 1906 published in Russian the article in which the concept of the Markov chain is proposed (Markov, 1910).

Concerning classification which is: "*...the process of assigning group membership labels to unlabelled observations. In this context, a group may be a class or a cluster...*" (McNicholas, 2016), the first statistical method is due to Ronald Fisher in his famous work on discriminant analysis. Lazarsfeld in 1950 proposed the first model-based clustering method. Metropolis (Metropolis et al., 1953) using the idea of the Markov chain developed an algorithm for Bayesian estimation of the model parameters. In 1962 with the arrival of electronic computation, Tukey's paper titled "The future of data analysis" argued for a more application- and computation-oriented discipline. The expectation-maximization algorithm which is undoubtedly the main tool to estimate many models under the maximum likelihood inference, was derived by Baum and colleagues in a series of papers in 1966 (Baum et al., 1970). In 1972 Cox proposed proportional hazards model (Cox, 1972) within the survival analysis contributing to the development of biostatistics. In 1979 Efron published a paper proposing an inferential method named bootstrap based on resampling from data, and during the same period, many Markov Chain Monte Carlo Methods were proposed as computational tools under the Bayesian framework.

In the 1988 finite mixture models were proposed to face the rise of many classification and clustering challenges as a consequence of the increasing availability of large and complex data (Everitt and Hand, 1981; McLachlan and Peel, 2000). The concepts of latent variables and causal inference are further developed (Bentler, 1980).

In 2016 data science seemed to represent a statistics discipline without parametric probability models or formal inference. As reported in Efron and Hastie (2021): "Data Science Association defines a practitioner as one who *uses scientific methods to liberate and create meaning from raw data*." In practice, the emphasis is on the algorithmic processing of large data sets for the extraction of useful information, with the prediction algorithms as examples. Data Science Association also wishes the following: "*A hopeful scenario for the future is one of an increasing overlap that puts data science on a solid footing while leading to a broader general formulation of statistical inference*." In a paper Persi Diaconis[1] published in *Statistical Science* an article[2] where at the end he stated that: "*Statistics is as healthy as it's ever been. One can see the prominence of machine learning, but they are really just using ideas that were developed in Statistics twenty or fifty years ago. They are applying them— that's great—but we are inventing the ideas that will be applied in the next twenty or fifty years. Statistics is a great field to be part of, and I'm still excited by it.*". "*It is the job of statistical inference to connect "dangling algorithms" to the central core of well-understood methodology. . . . Our optimistic scenario has the big-data/data-science prediction world rejoining the mainstream of statistical inference, to the benefit of both branches*."

A statistician approaches the study of collective phenomena with a scientific method through statistical science, which is useful for describing data and making inferences. Statistics developed in various scientific fields, leading to the emergence of numerous disciplines such as demography, econometrics, biometrics, anthropometrics, and business, among others. It faces the challenge of many applications involving large and complex data sets. Among them, environmental data and financial data constitute

[1] See the website: `https://en.wikipedia.org/wiki/Persi_Diaconis`

[2] Available at the following website: `https://projecteuclid.org/journals/statistical-science/volume-28/issue-2/Another-Conversation-with-Persi-Diaconis/10.1214/12-STS404.full`

some primary examples. The complexity of the vast range of new financial products continuously being introduced on the financial markets and the inherent uncertainty associated with stock prices, interest rates, and foreign exchange rates have given rise to the emergence of a new scientific field: mathematical finance. Many financial operations require decisions based on pre-defined rules, like option pricing or risk monitoring. High-frequency trading is a prime example; statistical and machine-learning algorithms are beneficial for this task.

Note that in 2020 there has been a World Statistics Day under the slogan: "Connecting the world with data we can trust". At the following website `https://worldstatisticsday.org/` you can find interesting speeches related to the event. Among them António Guterres (UN Secretary-General) said: "*Statistics are fundamental for evidence based policy making. Current, reliable, timely and trusty data help us to understand the changing world that we live and to drive the transformations that are needed leaving no one behind*".

1.1 Statistical inference

Statistical data analysis has various goals that involve the following aspects, among others:

1. Understanding data through summary statistics and visualizations;
2. Explaining the observed associations between variables;
3. Modelling one or more variables through the others;
4. Using the estimated models to make predictions over the phenomenon of interest;
5. Establishing causal claims.

In particular we focus on the construction of a statistical model formulated on the basis of certain assumptions. Laws are formulated using probabilistic theory and based on certain assumptions, and random variables represent how measured characteristics can vary from observation to observation. Statistical inference allows us to go back

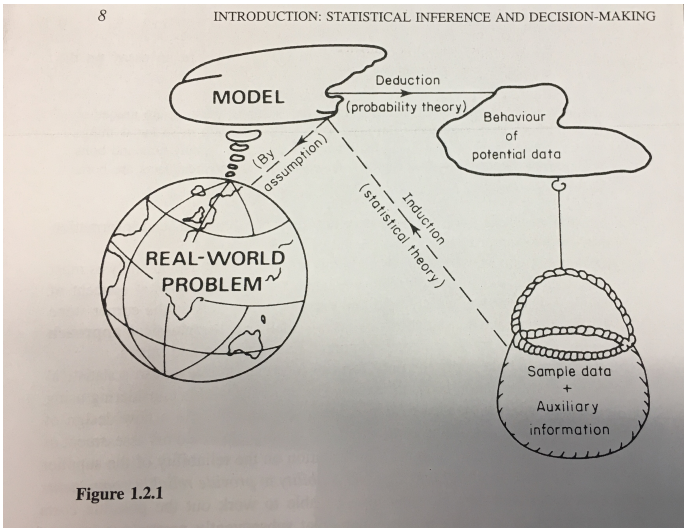


Figure 1: *Example of the deductive and inductive statistical reasoning. The practical situation, and information present in the observed data, links between probability theory and statistical models (figure taken from Barnett, 1975).*

from the description of a part (sample) to that of a larger whole and allows one to verify hypotheses formulated on the phenomena under study. Figure 1 shows the diagram of statistical inference proposed in Barnett (1975), which illustrates the deductive and inductive path of statistical science.

Probability is the foundation and the language for statistics, enabling many powerful methods for using data to learn about the world. Probability provides the formal framework for deduction: starting from an assumed model, it allows us to derive logical consequences about observable data. Statistical inference, instead, works in the opposite direction — it uses observed data to draw conclusions about the underlying population or model (induction). Understanding this distinction helps prevent common mistakes, such as confusing model assumptions with empirical evidence or interpreting probabilistic results as deterministic truths. Generally, the variability found in data is considered to be consistent with the expected variability in a random sample of observations assumed to be realizations from a supposed density (probability) distribution of an underlying random variable (James et al., 2013). Collecting appropriate data is one of the task involved in this process and it may be related to perform an experiment, look for observational data and scraping data elsewhere.

Models are formulated through suitable assumptions and parameters, referred to unknown quantities of the reference population, are estimated, with different methods, using sample data, i.e., a randomly chosen subset of the population. In such a way, general laws are validated for the entire reference population, after verifying the adequacy of the proposed model. Reliable estimations and predictions are provided by exploiting random variation. As George E. P. Box said: "All models are wrong, but some are useful", for more details on this quotation, see the following web site: `https://en.wikipedia.org/wiki/All_models_are_wrong`. In the study of collective phenomena it is necessary to:

- define the event of interest, and identify the collectivity in which the phenomenon occurs, e.g., cryptocurrency price and reference markets according to the market capitalization;

- choose the characteristics of the community considered of interest for understanding the phenomenon, e.g., the first year in which the currency operated digitally.

Furthermore, it is necessary to formulate hypotheses concerning the collectivity or a specific theory that one intends to be validated on the basis of the observed data.

Statistics is a science that, like few, considers uncertainty systematically and quantifies it. Therefore, its principles are used to validate theory in many sciences. Consider, for example, astrophysics, which operates with increasing precision in predominantly observational contexts, or through simulated experiments, mimicking with powerful tools situations akin to those assumed possible. At the European Centre for Nuclear Research (CERN), the conceptual framework of hypothesis testing developed since many years earlier (Neyman and Pearson, 1933) when there were small data sets and scientists with small experimental constraints, was adapted to the problem under study to the discovery of the Higgs Boson. A modern statistical method was applied to a problem as complex as the discovery of the Boson.

The statistical model mainly uses probability concepts like that of a random variable to provide a simplified description of the associations or causations postulated in a phenomenon under study. Statistical science, or statistical learning, provides both methods and models for the data analysis once the data have been collected through experimental or observational studies. Sir John Wilder Tukey was an American mathematician and statistician known for proposing data representation through the figure named box plot, which is employed for a graphical representation and description of the data through quartiles. He defined data analysis as the set 'of ways to plan data collection to make analysis easier, of procedures to analyze data, of theorems of mathematical statistics to interpret results and make procedures more precise and accurate' (Tukey, 1962).

Note that the current definition and distinction between statistical science and data science is somewhat nebulous. In fact, some define statistical science as the application of statistical methods to already sorted (cleaned) datasets and others broaden the definition (Wing, 2020; Peng and Parker, 2022). Statistics primarily focuses on inference, whereas data science often emphasizes predictive performance. In statistical science real-world and theoretical aspects are interconnected through data, statistical models, and algorithms. The theoretical aspects are concepts and probabilistic rules defined through random variables, models with parameters, and error components (or disturbances) with which it is possible to make inferences to draw conclusions (Tukey, 1962). The variability found in the data must be compatible with that expected according to determinations from random variables. Unobservable components contribute to the determination of the observed data; algorithms are used both for data analysis and to estimate the model parameters.

It should be noted that models are stochastic in nature, i.e., they assume random errors or disturbances. The resulting uncertainty allows probabilistic evaluations. For instance, suppose that in the study comparing two different incentives to invest in new capital goods for firms, the firms on the experimental incentive had an average profit gain of 100,000 Euros. What can we say about the average profit change if hypothetically all firms of the same sector benefit from this incentive? An inferential

statistical method provides an interval of numbers (confidence interval in classical inference and credibility interval in Bayesian inference) within which we can predict that the average profit change would increase. For example, the analysis might enable us to conclude, with a small probability of being wrong, that the average profit change for all firms in the same sector would fall between 85,000 and 115,000 Euros.

In summary, we can schematize the concepts we have outlined according to the following Figure 2 proposed in Kass (2011). The data generated in the real world may derive from observational or experimental studies, and what is observed in the data may also be the results of unobservable quantities. The data is characterized by regularity and variability, and the exploratory analysis of the data, sometimes using algorithms or statistical or machine learning techniques, leads to syntheses that are derived from models that are not probabilistic.

On the other hand, in the theoretical world, probabilistic rules are considered through the underlying random variables defining a statistical model along with random error disturbances. The parameters of the models are known quantities of interest in the population, which are estimated through algorithms, and the parameter estimates are presented along with their measure of uncertainty, thus leading to certain conclusions.

1.2 Review of some basic definitions

A statistical or sample unit is the subject of observation and is inherent to each phenomenon that constitutes the collective event of the study. For example, an experiment might have firms or banks as subjects if its goal is to compare profits yields.

Population represents that set of units identified as equal in the problem under study. The population is the entire set of subjects/things of interest. The collective may be finite or potentially infinite when it consists of many units whose amount is not precisely detectable.

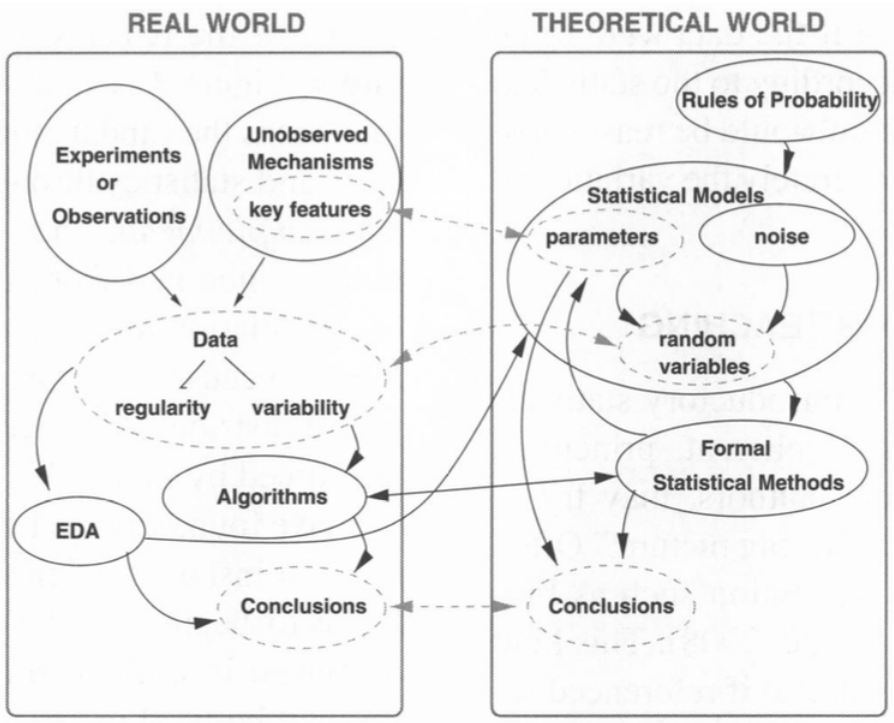


Figure 2: *A conceptual map of statistical inference as proposed in Kass (2011) (at Page 6).*

A sample is the set of subjects from the population for which data are available. The goal of most statistical data analyses is to learn about populations (to provide a general statement) based on data from a sample (this is the inductive process depicted in Figure 1). The main interest is related to the population from which the sample is derived. The sample is a part of the population about which information is available. We draw conclusions about populations based on information from samples. The ideal method of picking out a sample to study is called random selection. This procedure uses a random method such as a computer-generated list of random numbers to select a unit so that each unit in the population has an equal chance of being selected. In this way the sample elements are selected randomly from the population, independently of any characteristics, and the statistical sampling (or resampling) theory is able to account for the expected variation.

Probability is important since the main concepts in statistics are expressed in terms

of random variables and their related probability distributions. A variable is a characteristic that can vary in value among subjects in a sample or population. It defines the characteristics of the statistical unit of reference; each feature is present for a statistical unit in specific ways.

Values of variables of interest may be detected according to a specific instant of time, for example, a firm's default or the number of employees. Otherwise, they may refer to an interval of time, e.g., the closing price of stock taken each five minutes.

Some variables are always time invariant and may become identifiers of the unit, such as the place of birth for the individual or registered office of a company. Time-varying variables are those where time is the circumstance for which they are measured.

Continuous, quantitative, or numeric variables are defined according to numerical values representing different magnitudes. They derive from measurement processes and depend on the instruments' accuracy and the unit of measurement, e.g., annual income in CFH or tenths of a second. They can take an infinite continuum of possible real number values. Quantitative variables can also be discrete if they can take a set of distinct, separate values, such as the nonnegative integers, e.g., the number of products purchased on Amazon. They have integer numerical values also derived from counting operations. Statistical methods for continuous variables are used for quantitative variables that can take a very large number of values, regardless of whether they are theoretically continuous or discrete.

Qualitative, categorical, or nominal variables are with measurement scales made of categories. Categorical variables have two types of measurement scales. They can be ordinal or rank-ordered, such as the rating (score) assigned by the satisfaction assessment of a service or credit rating assigned by Standard & Poors to the Credit Suisse Group (downgraded to BBB from BBB+), or nominal in which the values are categories, such as the yes or no vote on a referendum, or preferred colors or whether employed (yes, no), their scale does not have a "high" or "low" end.

Some studies use a planned experiment to generate data. An experiment compares subjects on a response variable under different conditions. Those conditions, which are levels of an explanatory variable, are called treatments. The researcher specifies a plan for assigning subjects to the treatments, called the experimental design. Good experimental designs use randomization to determine which treatment or program a subject receives.

In many application areas, conducting experiments to answer the questions of interest is impossible. We cannot randomly assign subjects to the groups we want to compare, such gender, race, educational level, annual income, or usage of guns. When individual encounters were not assigned at random to the various situations or treatments we refer to this as observational studies. In these studies the outcomes are measured without any experimental control of the subjects. In such cases, we also refer to unobserved confounding since one can never rule out completely the possibility of influential unobserved variables.

The search for the causation or association of interest proceeds disjointedly. Association does not imply causation. An interesting graphical scheme is provided in Hernán and Robins (2019) and reproduced in Figure 3. The population is divided into two parts of different sizes: treated (white, e.g., companies that received incentives) or untreated (grey) units. Causation requires comparing units in the white and grey figure. On the other hand, it is easy to find associations with observational data, but those associations are often explained by other variables that may not have been measured in a study. In observational studies, the researcher often cannot intervene to fix the characteristics of the system in order to examine the outcome with respect to deterministic variations in certain factors but does have observations about facts, behaviors, and non-induced actions or situations. Therefore, analyzing data from these studies requires more stringent assumptions and specific statistical procedures. For further discussion, see, among others, Dawid (2002).

Causal inference can be conducted more quickly in the context of randomized studies as there is randomization, i.e., the random allocation of treatment to units, which allows

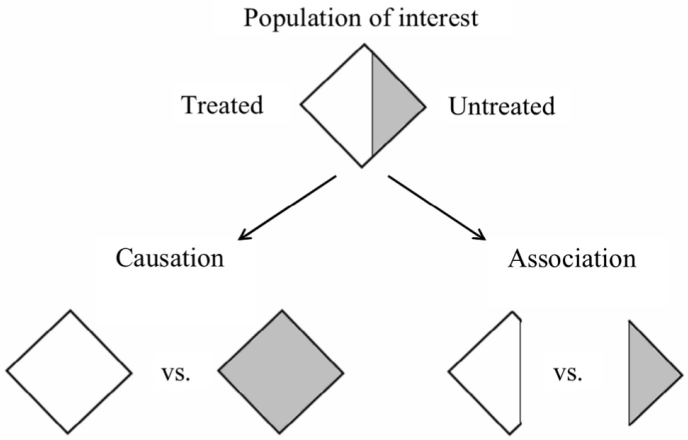


Figure 3: *Difference between association and causation. (From Hernán and Robins, 2019, Pag. 11).*

the effect to be detected. In such studies, the experimenter (e.g., a biologist) determines the decision on which units to treat randomly, e.g., based on the realization of an event such as the tossing of an unrigged coin. In the observational studies, stringent hypotheses are required to assess a causal effect. Therefore, if the scientific interest is to determine the causal effects between events, counterfactual theory, developed in the field of statistics (Rubin, 1974), may be employed to facilitate inductive reasoning providing a suitable conceptual framework to estimate causal effects (Holland, 1986).

2 A review of random variables, distributions and measures of association

In the following, a brief review of some basic concepts is presented. For more details, the student can refer to the book by Agresti and Kateri (2021). A more thorough treatment of basic probability and inferential concepts can be found in the book of Gentle, titled *Theory of Statistics*, freely available at the following webpage: `https://mason.gmu.edu/~jgentle/books/MathStat.pdf`.

A random variable is a mathematical abstraction that can serve a model for *observable quantities*. It is a function that assigns a numerical value to each point in the sample space. The total probability over all possible values equals 1. A probability distribution lists the possible outcomes of a random variable along with the associated probabilities. For a random phenomenon, the set of all the possible outcomes is called the sample space. This is the set of all samples of the same size that can be randomly drawn from a population.

We briefly recall the following probability rules. Let $P(A)$ denote the probability of an event A. All probabilities satisfy:

- $P(A) \geq 0$;
- denoting the sample space as S, $P(S) = 1$;
- if A and B are disjoint events (i.e., they contain no common outcomes), containing no common outcomes, $P(A \cup B) = P(A) + P(B)$
- the probability that an event does not occur is one minus the probability that it does occur $P(A^c) = 1 - P(A)$
- if A and B are not disjoint events, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- the probability that an event A given that an event B occurred is the conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Continuous random variables have an infinite continuum of possible values. Their probability distributions assign probabilities to intervals of real numbers rather than to individual values. The probabilities are determined by a probability density function. A continuous random variable is a random variable with a continuous distribution.

The function $f : \mathbb{R} \to \mathbb{R}$ is said to be a probability density function, if and only if

- $f \geq 0$,
- f is integrable,
- $\int_{-\infty}^{+\infty} f(x)dx = 1$.

Let X be a random variable. Its probability distribution can also be specified by F, its cumulative distribution function. The probability $P(X \leq x)$ that a random variable takes a value less than or equal to x is called the cumulative probability. The cumulative distribution function is

$$F(x) = P(X \leq x),$$

for all real numbers x.

Let f be a probability density function: for a continuous random variable X has density f if

$$F(x) = \int_{-\infty}^{x} f(t)dt.$$

The probability density function is integrated over all values up to x so that the cumulative distribution function is the accumulated area under the probability density function.

Cumulative distribution functions have the following properties. Let X be a random variable with cumulative distribution function $F_X(t) = P(X \leq t)$. Then

- for every $t \in \mathbb{R}$ we have $0 \leq F(t) \leq 1$,
- F is a non-decreasing function,
- $\lim_{t\to-\infty} F(t) = 0$, $\lim_{t\to+\infty} F(t) = 1$,

- F is right-continuous.

The area under the function over an interval of values, which equals its integral over that interval, is the probability that the random variable falls in that interval.

One can characterize a probability distribution by dividing points, which are called *percentiles*. The $(100p)$th percentile, $0 < p < 1$, is a point π_p such that

$$P(X \leq \pi_p) = p \qquad \text{and} \qquad P(X > \pi_p) = 1 - p$$

So, π_p is the solution of the equation $F(\pi_p) = p$.

The most important percentiles are the median (a point such that at most, half the population have values less than the median, and, at most, half have values greater than the median), $m = \pi_{1/2}$, and the quartiles, $q_1 = \pi_{1/4}$ and $q_3 = \pi_{3/4}$ (called the first and third quartiles, respectively).

A continuous random variable can take any value in an interval although the probability that it is equal any particular value is exactly 0. The cumulative density function is differentiable and the derivative is called the probability density function. Probability is represented by area under the probability density curve *not* by the value of the probability density function at a point. Once the probability density function is integrated we get a probability.

In the following we review the Gaussian (or Normal), and Student-t distributions.

2.1 Gaussian distribution

In the following we consider the Gaussian distribution (also named *Normal* distribution). Gaussian distribution (also named *Normal* distribution) proposed by Gauss[3] has a simple mathematical formulation. It is extremely widely used in statistics because of the *central limit theorem*, which says that under very weak assumptions, the sum of a large number

[3]see `https://en.wikipedia.org/wiki/Carl_Friedrich_Gauss`

of independent and identically distributed random variables has an approximately Normal distribution, regardless of the distribution of the individual random variables.

A continuous random variable Y is said to have a Normal distribution and it is denoted as

$$Y \sim N(\mu, \sigma^2),$$

if its probability density function is given by

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(y-\mu)^2 \right),$$

where $-\infty < y < \infty$, $\mu \in \mathrm{R}$, $\sigma^2 > 0$. The values on the x-axis corresponding to a negligible density lie after the extremes $\mu \pm 3\sigma$.

A continuous random variable Z is said to have the *standard Normal distribution* if its probability density function is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2 \right),$$

and we write $Z \sim N(0,1)$. Note that the constant $\frac{1}{\sqrt{2\pi}}$ is called normalizing constant since it is needed to make the probability density function to integrate to 1.

The standard normal cumulative density function is the accumulated area under the probability density function and it is the following

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}t^2 \right) dt.$$

From the standard normal we deduce: i) symmetry of the probability density function; ii) symmetry of tail areas $\Phi(z) = 1 - \Phi(-z)$; and iii) symmetry of Z and $-Z$.

The following Figures 4, 5, and 6 depict in blue the whole or part of the area of the density function.

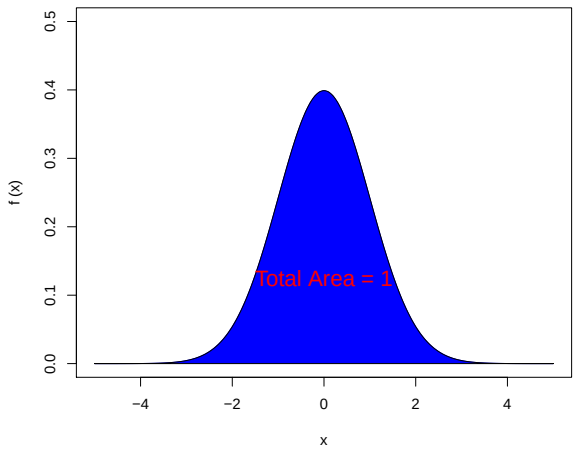


Figure 4: *Plot of the density function of the standard normal distribution*

2.2 Student-t

The random variable X has a Student-t distribution with ν degrees of freedom, denoted as $X \sim T_\nu$, if its density function is given by

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}.$$

In this case notice that:

- the shape of $f(\cdot)$ is similar to that of a standard normal random variable. However, $f(\cdot)$ has fatter tails compared to the ones of a normal distribution: this makes the Student-t distribution especially useful for modeling phenomena in which the extreme values play a non negligible role, for example *log-returns of stocks showing high volatility*.

- for $\nu \to \infty$, $f(x)$ approaches the density function of a normal random variable for every $x \in \mathbb{R}$.

If $Z \sim \mathcal{N}(0,1)$, $U \sim \chi^2_\nu$ and Z and U are independent, then

$$\frac{Z}{\sqrt{U/\nu}} \sim T_\nu$$

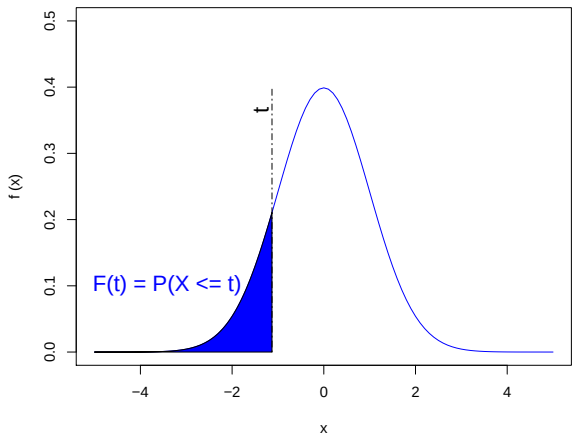


Figure 5: *Value of the integral* $\int_{-\infty}^{t} xf(x)dx$ *of the standard normal distribution in blue*

.

Let $\bar{X}_n = \sum_{i=1}^{n} X_i/n$ the sample mean and $S^2 = \sum_{i=1}^{n}(X_i - \bar{X}_n)^2/(n-1)$ the unbiased sample variance. It is possible to show that

- $Z = (\bar{X}_n - \mu)/(\sigma/\sqrt{n}) \sim \mathcal{N}(0,1)$;
- $U = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$;
- Z and U are independent.

The following Figure 7 recalls some properties of the expected value and variance.

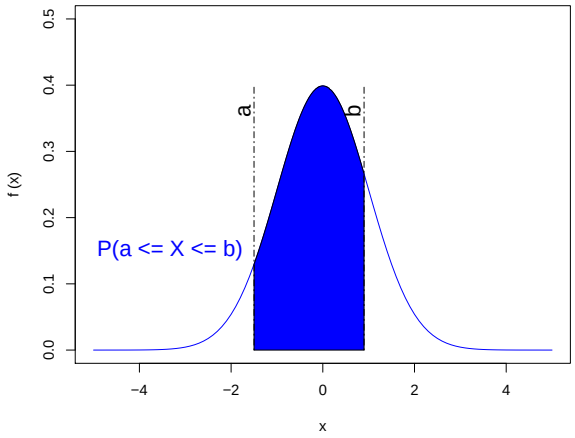


Figure 6: *Area between values a and b of the standard normal distribution in blue*

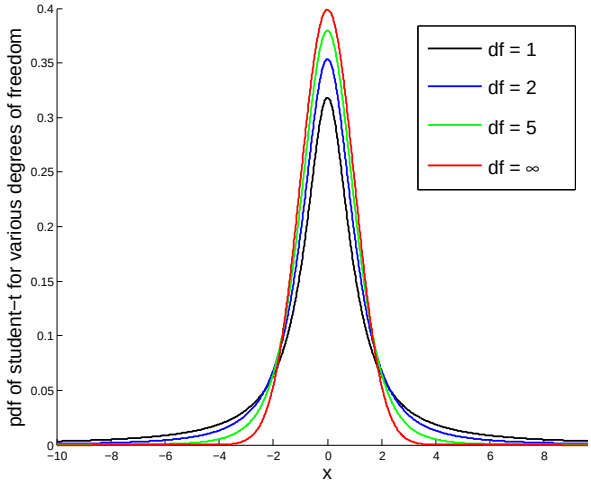


Figure 7: *Shapes of the Student-t distribution for increasing values of the degrees of freedom*

2.3 Empirical cumulative distribution function

The empirical cumulative function is defined as the fraction of data smaller than or equal to x. Given a random sample of size n from a random variable X with cumulative distribution function $F(x)$ and given ordered n sample values $x_i \leq x_{i+1}$, $i = 1, \ldots, n-1$, is called empirical cumulative distribution function the function $\hat{F}_n$ with values in the interval $[0, 1]$ which assigns to each x its sample weight $1/n$:

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x_i),$$

where $I(.)$ is the indicator function, $I(X_i \leq x_i) = 1$ if $X_i \leq x_i$ and $I(X_i \leq x_i) = 0$ otherwise. Therefore, $\hat{F}_n(x)$ is the sample proportion of the n observations that fall at or below x.

In the situation of total absence of information about the law of probability F of a random variable from which the sample is drawn, the empirical cumulative distribution function can be assumed as good quality estimate of $F(x)$ since due to the Glivenko-Cantelli theorem it strongly converges strongly and uniformly to the true cumulative distribution function.

2.4 Quantile-Quantile plot

The QQ plot (quantile-quantile plot) is a graphical comparison of observed data with a theoretical distribution. It is the plot the k-th smallest observation against the expected value of the k-th smallest observation out of n in a standard normal distribution. The point is that in this way you would expect to obtain a straight line if data come from a normal distribution with any mean and standard deviation. It plots the theoretical quantiles $F^{-1}(p)$ (of the Gauss distribution) for appropriate values of the proportions (p) on the x-axis and empirical quantiles $\hat{F}^{-1}(p)$ on the y-axis. If the empirical values align on the bisector of the first quadrant, i.e. the graph follows a straight line $y = x$ having intercept 0 and slope 1, then the empirical values are in line with the theoretical ones.

Otherwise, if there are deviations from the straight line, the two distributions are not the same, that is the observed data distribution differ from the reference distribution.

For n observations $\{x_i\}$ let $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ denote their ordered values, called order statistics. Let q_i be the $i/(n+1)$ quantile of the standard normal distribution, for $i = 1, \ldots, n$. When $\{x_i\}$ are a random sample from a normal distribution, the plot of the points $(q_{\frac{1}{n+1}}, x_{(1)}), \ldots, (q_{\frac{n}{n+1}}, x_{(n)})$ where $q_{\frac{i}{n+1}}$ is the ordered quantile, should approximately follow a straight line, especially when n is large. This is a normal quantile plot.

2.5 Discrete random variables: Expected value and variability

A discrete random variable takes on values in a countable set. The outcomes are the distinct values the random variable can assume, usually integers.

Let $P(x)$ denote the probability that the discrete random variable X takes value x. This function is called the probability mass function and gives the probabilities for the possible outcomes of a random variable.

Then,

$$0 \leq P(x) \leq 1 \ \ and \ \ \sum_{all \ \ x} P(x) = 1,$$

where the sum is over all the possible values of X.

The expected value of a discrete random variable X is

$$\mathbb{E}(X) = \sum_x xP(X),$$

where the sum taken over all possible values x of X.

The variance is denoted by σ^2, and it is defined as

$$\sigma^2 = \mathbb{E}(X - \mu)^2 = \sum_x (x - \mu)^2 P(x),$$

which represent the average square deviation of X from the mean, with the sum taken over all possible values of x.

2.6 Joint and conditional distributions

Random variables are used as models for observable events, and, in many cases, most statistical analyses involve two or more more random variables simultaneously. It is often necessary to consider multivariate data, that is, data consisting of multiple related variables or measurements in a given applied context. The joint analysis of a series of measurements can help us identify patterns and trends that may be used to forecast future observations. For example, when considering cryptocurrencies (digital currencies), the rates of returns of BTC (Bitcoin) and ETH (Ethereum) appear to be related. At any time t, if the BTC return is significant, the ETH return is more likely to be relatively large.

A joint (or multivariate) probability distribution specifies probabilities for all the possible combinations of values of the random variables, thus capturing how multiple random variables interact. For simplicity, we first focus on the bivariate case involving two random variables, X and Y.

Given two discrete random variables X and Y, defined on the same probability space $(\Omega, \mathcal{A}, P)$, the *joint density mass function* of X and Y, denoted by $f(x, y)$, is defined as

$$f(x, y) = P(X = x, Y = y).$$

It satisfies

$$f(x, y) \geq 0 \ \ and \ \ \sum_x \sum_y f(x, y) = 1,$$

where the sums are taken over all possible values of X and Y. The joint distribution can be represented graphically over the plane of possible (x, y) values, see Figure 8.

The probability distribution of a single random variable, considered by itself without reference to any other random variable, is called a marginal distribution. A joint probability function determines each *marginal probability distribution* by summing (in the discrete case) or integrating (in the continuos case) over the possible values of the other

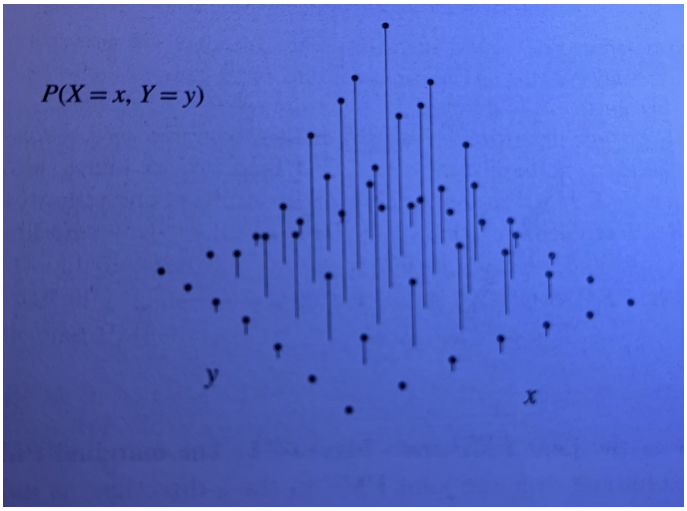


Figure 8: *Joint probability mass function of discrete random variables X and Y (figure taken from Blitzstein and Hwang (2015))*

variable. It follows that

$$P(X = x) = \sum_y p(x, y),$$

where the sum involves all the possible values y in support of Y. Figure 9 shows the marginal probability of two discrete random variables.

A *conditional probability distribution* specifies the probabilities of the outcomes of one random variable given the value of another random variable. In other words, it describes the distribution of one random variable conditional on a particular value of another random variable.

In the discrete case, the the conditional distribution can be obtained by applying the conditional probability rule

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)},$$

for $P(Y = y) > 0$, and similarly

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)},$$

for $P(X = x) > 0$.

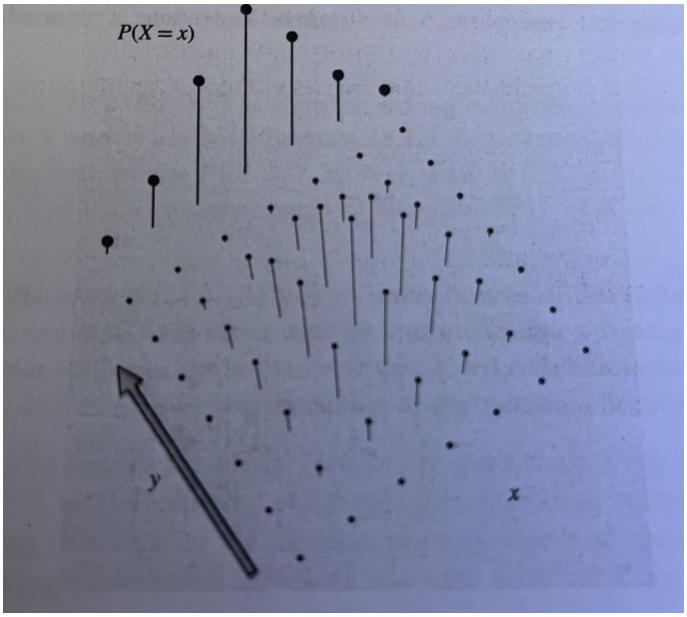


Figure 9: *Marginal probability obtained by summing over the joint (indicated by the arrow) probability of discrete random variables X and Y (from Blitzstein and Hwang (2015))*

For continuous random variable Y, the conditional expectation of Y at a particular value x of X is its mean

$$\mathbb{E}(Y|X = x) = \int_y yf(y|x)dy.$$

A useful result, called the law of iterated expectation, says that

$$\mathbb{E}(Y) = \mathbb{E}[\mathbb{E}(Y|X)].$$

If the random variables X and Y are independent, the value taken on by one of the random variables does not affect the probability distribution of the other one. Two random variables X and Y with marginal probability functions $f_1(x)$ and $f_2(y)$ are independent if

$$f(x|y) = f_1(x) \ \ and \ \ f(y|x) = f_2(y),$$

for all possible values x of X and y of Y. Then, their joint probability function $f(x, y)$ satisfies

$$f(x, y) = f_1(x)f_2(y) \ \ for \ \ all \ \ x \ \ and \ \ y.$$

In general, we observe a finite random sample $y_1, y_2, \ldots, y_n$ from a population, which are interpreted as realizations of a random variable Y, assumed in the present discussion to be absolutely continuous. In the case of multiple random variables $Y_1, Y_2, \ldots, Y_n$ they are said to be mutually independent if their joint probability distribution factors as the product of the marginal probability distributions:

$$f(y_1, \ldots, y_n) = f_1(y_1) \cdots f_n(y_n),$$

for all possible values $(y_1, \ldots, y_n)$. This property implies that the variables are pairwise independent, that is, $f(y_i, y_j) = f_i(y_i) f_j(y_j)$ for all $i \neq j$. However, it is important to note that pairwise independence does not imply mutual independence.

2.7 Measures of associations

To study associations among random variables, we consider some direct measures of linear association among two variables, such as the covariance between pairs of variables. Covariance measures the tendency of two random variables to increase or decrease together relative to their expected values: positive covariance between X and Y indicates that when X is above its expected value, Y also tends to be above its expected value, while negative covariance indicates that when X is above its expected value, Y tends to be below its expected value.

- The covariance between two random variables Y and X is defined as

$$Cov(Y, X) = \sigma_{YX} = \mathbb{E}[(Y - \mathbb{E}[Y])(X - \mathbb{E}[X])],$$

  and

$$Cov(Y, X) = \mathbb{E}(YX) - \mathbb{E}(Y)\mathbb{E}(X).$$

The covariance can take any real value. Its magnitude depends on the units of measurement. If Y and X are independent, then $Cov(Y, X) = 0$, and we say that random variables with zero covariance are uncorrelated. Note that X and Y being uncorrelated

does not imply that they are independent.

A similar measure that is free of the units of measurement is obtained by dividing covariance by the corresponding standard deviations of the two variables. The linear correlation between random variables X and Y having finite variance is

$$Cor(Y,X) = \rho_{YX} = \frac{\sigma_{YX}}{\sigma_Y \sigma_X} = \frac{Cov(Y,X)}{\sqrt{V(Y)V(X)}}.$$

The correlation coefficient takes values in the interval $[-1,1]$. It measures the strength of the linear association between two variables: a large absolute value indicates a strong linear association. However, the converse is not true: a strong association between two variables does not necessarily imply a large value of the covariance or correlation coefficient.

For multivariate distribution with more than two random variables, we use the variance-covariance matrix denoted as $\mathbf{\Sigma}$ and the correlation matrix denoted as $\boldsymbol{R}$. These matrices are symmetric and nonnegative definite. The variance-covariance matrix of the random variables $Y_1, .., Y_n$ is defined by

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{12} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1n} & \sigma_{2n} & \cdots & \sigma_n^2 \end{bmatrix}$$

where $\sigma_i^2 = \text{Var}(Y_i)$ and $\sigma_{ij} = Cov(Y_i, Y_j)$, $i \neq j$. This is a squared symmetric matrix (since $\sigma_{ij} = \sigma_{ji}$) and is positive semidefinite.

The correlation matrix of $Y_1, .., Y_n$ is defined as

$$\boldsymbol{R} = \begin{bmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{12} & 1 & \cdots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1n} & \rho_{2n} & \cdots & 1 \end{bmatrix}$$

where ρ_{ij} is the correlation between Y_i and Y_j (ρ_{ii}=1). Also this matrix is squared, symmetric and positive semidefinite. [** For matrix algebra see the file provided in the elearning page of the course titled: *An introduction to matrix algebra*].

The sample covariance calculated from an observed sample is

$$s_{yx} = \sum_{i=1}^{n}(y_i - \bar{y})(x_i - \bar{x})/(n-1)$$

and the sample correlation is

$$r_{yx} = \frac{s_{yx}}{s_x s_y}.$$

The index of the association between two quantitative variables is also named coefficient of Bravais-Pearson correlation. It is a symmetric index: it does not distinguish between response or explanatory variables. A positive sign indicates a positive correlation: the two variables tend to increase or decrease in the same direction. We speak of a perfect linear association between the two variables when the correlation is equal to 1. A zero-value of the correlation implies that as one variable increases or decreases, no increasing or decreasing trend is associated with the other variable and vice versa. A value of 0.85 referring, for example, to the correlation calculated between the value estimated for a property by a real estate agency and the overall surface area of the same, indicates a strong linear correlation between the two variables, i.e., when there is an increase in surface area there is an increase in value, and vice versa. Another example is the observed correlation of the daily closes of the Dow Jones and S&P 500 stock market indexes which, for a certain period, was 0.993.

Note that two random variables may be dependent but not linearly correlated: the correlation coefficient maybe equal to zero, even though there is a nonlinear (for example quadratic) dependence between the two variables. Figure 10 illustrates this case, where Y is a deterministic function of X, yet X and Y are uncorrelated. Suppose that the x-axis shows the observed values of the logarithm of the return of a share and the y-axis shows the values of an investment fund. One can observe negative association for values of X below zero and a positive association for values of X above

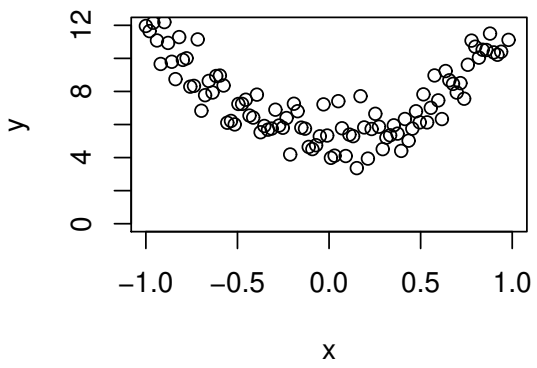


Figure 10: *Draws from the joint distribution of* (X, Y) *under a certain dependence structures*

zero. The value of the linear correlation is misleading in the presence of spurious (or masked) associations (Lazarsfeld, 1955). This occurs when the association between two variables depends on a third variable that, if omitted, changes or eliminates the observed association between the first two variables. For example, consider a context in which a particularly high positive correlation is found between height and individual income. If income is positively correlated with height this may reflect an indirect association: income depends, as in Italy, on gender, and height is also positively associated with gender. Therefore, the observed association between height and income may be due to their common association with gender. Remember the need to very the link with other potentially associated variables[4]. "It serves as a valuable reminder of the importance of careful statistical analysis and the potential pitfalls of simplistic interpretations of data" (from Wikipedia).

2.8 Partial linear correlation coefficient

An important measure is that provided by the coefficient of partial correlation, which quantifies the linear association between two variables adjusted for the influence of

[4]This is known as Simpson's paradox, see the following website `https://en.wikipedia.org/wiki/Simpson%27s_paradox` for some additional examples.

other variables. The coefficient of partial correlation between two variables (x_1 and x_2) net of a third x_3 is denoted as

$$\rho_{yx.z} = cor(y, x \quad net \quad z),$$

while the sample coefficient is denoted

$$r_{yx.z} = cor(y, x \qquad net \quad z),$$

and represents the correlation observed in the data between y and x net of z.

Note that it can be expressed according the bivariate correlations as in the following example considering the partial correlation between x and y net to z

$$r_{xy.z} = corr(x, y|z) = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}.$$

Therefore only in the case of null correlation between x and z and between y and z the partial correlation is equal to the standard correlation $r_{xy.z} = r_{xy}$.

It ranges from $[-1, 1]$; the extreme values represent perfect discordance (if equal to -1) or perfect linear concordance (if equal to 1) between the two variables, holding the third variable z. The variables x_1 and x_2 are concordant given of x_3 if $r_{12.3} > 0$ and discordant given x_3 if $r_{12.3} < 0$. In the latter case, the two variables tend to vary in opposite directions: when one increases, the other tends to decrease, conditional on the third variable. This concept can be generalized to more than one control variable, so that it measures the correlation between x and y given $z, w, \ldots$.

2.8.1 Scatterplot to visualize bivariate associations

The bivariate scatterplot presented in Figure 10 shows the observed values of two variables in the plane. In the Cartesian plane, each point has coordinates corresponding to the value of the variable placed on the x-axis and the value of the variable placed on the y-axis. The distribution of the points may reveal a linear trend or other patterns,

and may also highlight the presence of groups of observations.

The scatterplot (or pairs plot) presented in Figure 3 allows the values of two traits measured for all statistical units to be displayed on the plane. The arrangement of the points may reveal linear trends or particular patterns, and the degree of dispersion of the points in the plane. When several variables are observed, the associations between pairs of variables can be displayed in a squared array. A scatter plot matrix is often used to represent jointly the associations between each pairs of variables.

This graphical representation is also useful for checking outliers and examining how the points are distributed in the plane. This issue will be discussed further below, but it is important to emphasize that that statistical methods based on the assumption of Gaussian distribution can strongly affected by outliers.

The scatterplot matrix is a square matrix wh all the graphs representing the diagrams referring to each pair of variables. It shows how the points are arranged in the plane, the connections between them, the trend of the observed values, and also which values are particularly distant from the center.

2.9 Bivariate and multivariate Gaussian distribution

2.9.1 Bivariate Gaussian distribution

Consider the bivariate random variable (X, Y) and let (x, y) be its realisation, $f(x, y)$ be the joint density function. Given constant values such that $a < b$ and $c < d$ the probability $\mathbb{P}(a < X < b, c < Y < d)$ is defined by the following volume:

$$\mathbb{P}(a < X < b, c < Y < d) = \int_a^b \int_c^d f(x, y) dy \, dx.$$

The joint density of (X, Y) follows a bivariate Gaussian (or Normal) distribution if the random variable resulting from every possible linear combination of the following type $W = aX + bY$, $\forall\, a,\ b \in \mathbb{R}$ has a univariate Gaussian distribution.

Parameters of this joint distribution are:

- the means μ_x, μ_y,

- the variances σ_x^2, σ_y^2,
- the correlation coefficient ρ_{xy} (recall that $cov(X,Y) = \rho_{xy}\sigma_x\sigma_y$).

The notation for the bivariate distribution is the following:

$$(X,Y) \sim N(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho_{xy}),$$

and the probability density function is

$$f(X=x, Y=y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho_{xy}^2}} \exp(-g),$$

where

$$g = \frac{\frac{(x-\mu_x)^2}{2\sigma_x^2} + \frac{(y-\mu_y)^2}{2\sigma_y^2} + \rho[\frac{(x-\mu_x)}{\sigma_x}\frac{(y-\mu_y)}{\sigma_y}]}{1-\rho_{xy}^2},$$

for $-\infty < x,\ y < +\infty$, $-\infty < \mu_x, \mu_y < +\infty$, $\sigma_x > 0$, $\sigma_y > 0$ and $-1 \leq \rho_{xy} \leq 1$.

Figure 11 shows the shape of the bivariate density function which appears to be bell-shaped in the three dimensional space with parameters taking the following values

$$(X,Y) \sim N\left(\mu_x = 0, \mu_y = 0, \sigma_x = 3, \sigma_y = 3, \rho_{xy} = -\frac{2}{3}\right) \sim N\left(0, 0, 3, 3, -\frac{2}{3}\right).$$

Some main properties of the distribution are listed below:

- The variables, if they are uncorrelated, are also independent (a property that does not hold in general for the other distributions), and the marginal distributions are Gaussian. In fact, if (X,Y) has a bivariate Gaussian distribution with uncorrelated variables $(X,Y) \sim N(\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho_{xy} = 0)$ then X and Y are independent:

$$f\left(x,\ y;\ \rho_{x,y} = 0\right) = f\left(x\right) f\left(y\right).$$

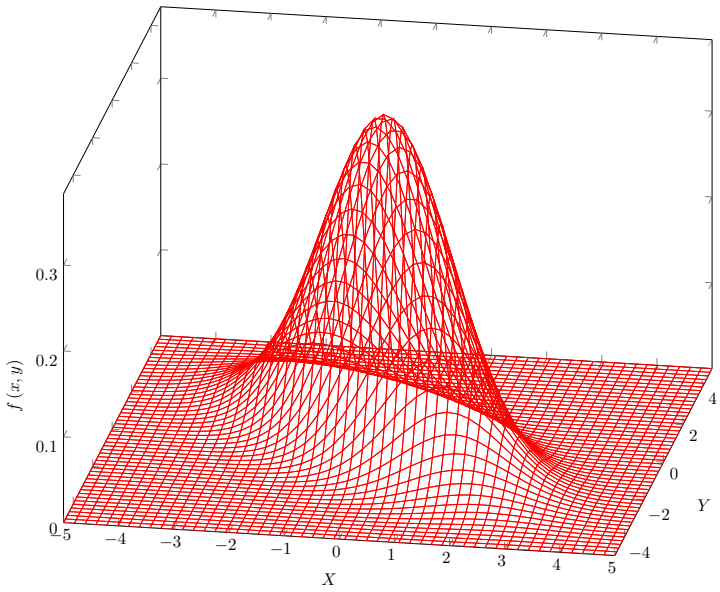


Figure 11: *Probability density function of the bivariate Gaussian distribution* $(X, Y) \sim N(0, 0, 3, -\frac{2}{3})$

- Marginalising with respect to X or Y, it results that $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$.

- The conditional distributions $(X|Y = y)$ and $(Y|X = x)$ are Gaussian distributions. In particular, it can be shown that:

$$\mathbb{E}[X|Y = y] = \mu_x + \frac{\rho_{x,y}\sigma_x\sigma_y}{\sigma_y^2}(y - \mu_y) = \mu_x + \rho_{x,y}\frac{\sigma_x}{\sigma_y}(y - \mu_y),$$
$$\mathbb{E}[Y|X = x] = \mu_y + \frac{\rho_{x,y}\sigma_x\sigma_y}{\sigma_x^2}(x - \mu_x) = \mu_y + \rho_{x,y}\frac{\sigma_y}{\sigma_x}(x - \mu_x),$$

  and

$$\text{Var}(X|Y = y) = \sigma_x^2\left(1 - \rho_{x,y}^2\right),$$
$$\text{Var}(Y|X = x) = \sigma_y^2\left(1 - \rho_{x,y}^2\right).$$

- The contour lines of a bivariate Gaussian distribution are ellipses; the correlation coefficient ρ_{xy} defines the direction of the major axis of the ellipse. Examples of contour plots for three bivariate distributions with different parameter values are shown in Figure 12. A contour plot is a two dimensional figure showing one dimensional curves on which the plotted quantity density f is a constant. The curves are defined by $f_j = f(x, y)$, $j = 1, \ldots, Nc$, where N_c is the number of contours that are plotted.

2.9.2 Multivariate Gaussian distribution

The multivariate Gaussian distribution generalizes the bivariate distribution introduced earlier. Consider the random vector $\boldsymbol{Y} = (Y_1, \ldots, Y_p) \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where $\boldsymbol{\mu} \in \mathbb{R}^p$ is the vector of averages and $\boldsymbol{\Sigma} \in \mathbf{R}^{p\times p}$ is the positive definite symmetric variance-covariance

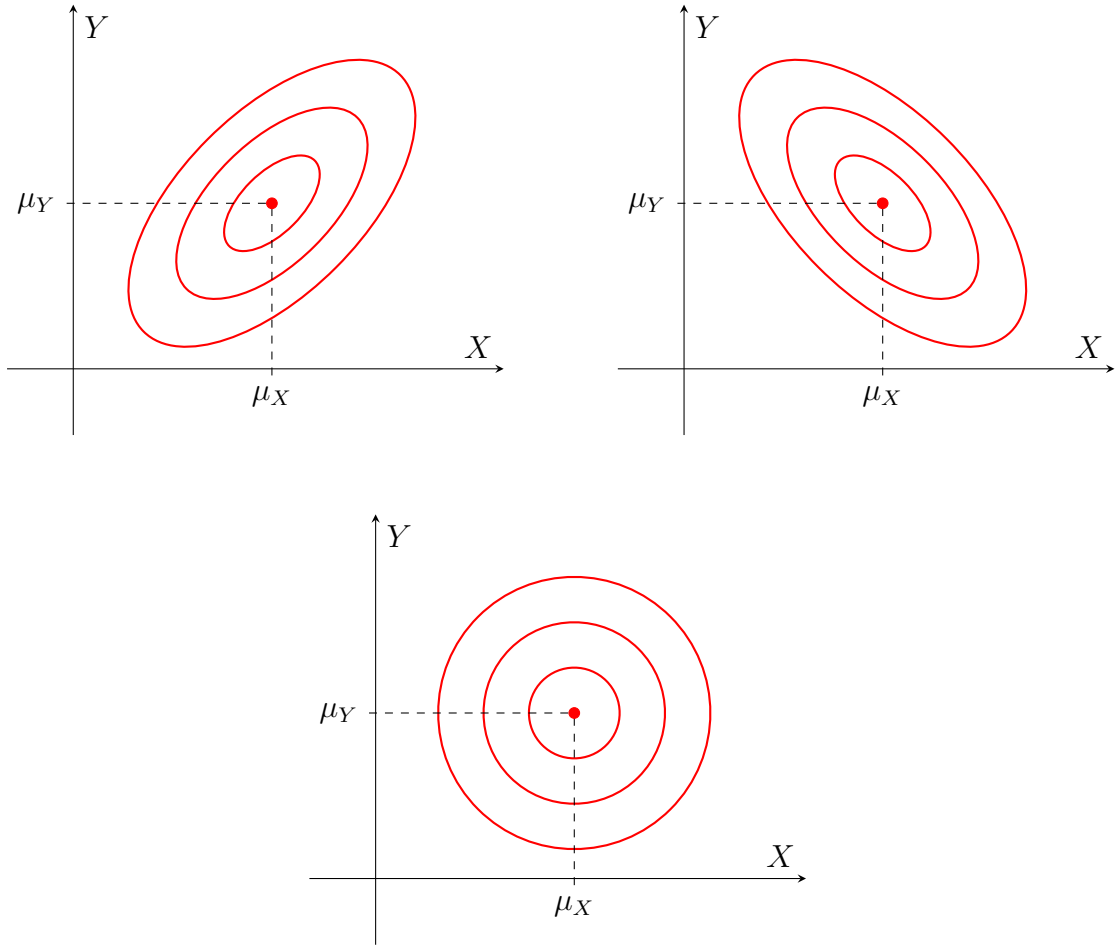


Figure 12: *Contour plots referring to the bivariate Gaussian distribution with different parameter values with positive, negative and zero correlation, respectively*

matrix.

The density function is as follows

$$f(\boldsymbol{y}) = f(y_1, \ldots, y_p) = \frac{1}{\sqrt{(2\pi)^p \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})\right),$$

The standard multivariate Gaussian distribution is written as $\boldsymbol{Z} \sim N_p(\boldsymbol{0}, \boldsymbol{I}_p)$, meaning that the means are null and the variance-covariance matrix is an identity matrix. Its density function is as follows

$$f(\boldsymbol{z}) = \frac{1}{\sqrt{(2\pi)^p}} \exp\left(-\frac{1}{2}\boldsymbol{z}^\intercal \boldsymbol{z}\right).$$

2.9.3 Singular value decomposition

A square real matrix $\boldsymbol{Q}$ is said to be *orthonormal* if its columns form an orthonormal set, meaning that each column has unit norm and the inner product between any two distinct columns is zero. In this case, the following holds:

$$\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{I},$$
$$\boldsymbol{Q}^T = \boldsymbol{Q}^{-1},$$

where $\boldsymbol{Q}^T$ and $\boldsymbol{Q}^{-1}$ are the transpose and the inverse of $\boldsymbol{Q}$, respectively, and $\boldsymbol{I}$ is the identity matrix.

Let $\boldsymbol{A} \in \mathbb{R}^{p\times p}$ be a real, symmetric, and positive-definite matrix. The *spectral theorem* states that $\boldsymbol{A}$ can be expressed as

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T,$$

where:

- $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_p)$ is a diagonal matrix whose entries are the eigenvalues of $\boldsymbol{A}$,
- $\boldsymbol{Q}$ is an orthonormal matrix whose columns are the corresponding eigenvectors of $\boldsymbol{A}$.

The above expression can obviously be further decomposed as the following weighted sum:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^T = \sum_{j=1}^{p} \lambda_j \boldsymbol{Q}_j \boldsymbol{Q}_j^T,$$

where $\boldsymbol{Q}_j$ is the j-th column of $\boldsymbol{Q}$.

By construction, these are ellipsoids, where the $\boldsymbol{Q}_j$ term provides the direction of the j-th component, while the λ_j term determines its magnitude. In particular, if applied to the covariance matrix $\boldsymbol{\Sigma}$, the spectral decomposition theorem provides a way to rank different types of associations among variables.

3 Statistical inference: Estimation

A mathematical model is a simple approximation for the assumed association among a set of variables in the population. A model uses a framework that incorporates assumptions about the random variability in observations of those variables and imposes a structure for describing and making inferences. Sampling distributions and statistical inferences are derived under a certain assumed model structure. As explained by Sir David Cox in Cox (1995): "*The construction of idealized representations that capture important stable aspects of a certain system is a vital part of general scientific analysis*", and as stated in Hennig (2010): "*Mathematical modelling reduces complexity and can make clearer and simpler perception of the reality possible. Models can provide decision support by generating comparable consequences from models formalising different decisions. Mathematical models often have surprising implications and give us a new, different view of the modelled phenomena*".

3.1 Review of the properties of estimators

We recall some basic properties of an estimator T. Generally these are **finite sample** and **asymptotic** properties.

Finite sample properties: Properties that are defined and can be evaluated for a sample size n which is fixed.

Asymptotic properties: Properties regarding behaviour of estimator for $n \to \infty$, i.e. regarding what happens if the sample gets larger and larger.

The three basic properties are the following:

- An estimator T is called unbiased if its sampling distribution centers around the parameter θ

$$\mathbb{E}_\theta(T) = \theta.$$

  In other words, T can be expected to yield θ on average.

If $E_\theta(T) \neq \theta$ for at least some θ, T is called **biased**. The **bias** of T is defined as

$$B_\theta(T) = E_\theta(T) - \theta.$$

Note that for a biased estimator a desirable asymptotic property is that its expected value tends to get closer to θ as the sample size n increases.

The variability of the estimator with respect to the parameter is the mean squared error (MSE) defined as

$$\text{MSE}(T) = \mathbb{E}(T-\theta)^2 = V(T) + B_\theta(T)^2,$$

and if the estimator is unbiased the MSE is its variance that is

$$V(T) = \mathbb{E}[T - \mathbb{E}(T)]^2.$$

So, the MSE incorporates two components, one measuring the variability of the estimator (opposite of precision), and the other measuring the bias (opposite of accuracy). Of the unbiased estimators, the one with smallest MSE is a minimum variance unbiased estimator. An estimator with minimum variance equivalently has minimum standard error.

- An estimator T is said to be consistent if it converges in probability to θ. This is an important property since the estimator is consistent if its sample distribution, as the total number of observations n increases (that is $n \to \infty$), tends to focus more and more on the actual value of the parameter. Note that this is a property that occurs in the limit of n. Consistency means that the estimator converges to the 'correct' value as $n \to \infty$. This is a quite fundamental property; inconsistent estimators are usually ruled out wherever consistent ones are known.

- An efficient estimator tends to fall closer to θ, on the average, then other estimators.

3.2 Likelihood function and maximum likelihood estimation

Under the assumption of a particular family of probability distributions, maximum likelihood (ML) estimators are consistent, asymptotically unbiased, efficient, and asymptotically normal. Note that the point estimate is the parameter value for which the observed data are most likely to occur.

The likelihood function (denoted as L), or its logarithm (the log-likelihood, denoted as ℓ) is not a probability distribution. Rather, it describes the support that the observed data provide for different parameter values. Due to its properties, the maximum likelihood method is one of the most widely used technique for deriving estimators, as it allows us to select the parameter values that are most plausible in light of the observed data. Therefore, the maximum likelihood estimate is the one that is best supported by the data under the assumed model.

We now recall some basic properties of the likelihood function, starting with the case of a single parameter. Let $\boldsymbol{x} = (x_1, \cdots, x_n)$ be an observed sample of n *independent and identically distributed observations* drawn from a population, and let $f(\boldsymbol{x}; \theta)$ denote their joint probability (or density) function, defined for a parameter $\theta \in \Theta \subseteq \mathbb{R}$. The *likelihood function* is

$$L(\theta) = f(x_1; \theta) \cdot \cdots f(x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta).$$

A plot of the likelihood function shows how plausible different parameter values are given the observed data; however, it does not integrate to 1 and therefore is not a probability distribution.

In practice, $L(\theta)$ provides the evidence in the data in favor of any given value of θ in Θ. When, for two values of θ, say $\theta^{(1)}$ and $\theta^{(2)}$, we have $L\left(\theta^{(1)}\right) > L\left(\theta^{(2)}\right)$, the observed sample is more likely under $\theta^{(1)}$; therefore, there is more evidence in favor of this value of θ. The likelihood function allows us to rank different values of θ according to the "*degree of support*" they receive from the observed data. Maximum likelihood estimation proposes an estimate of θ (when it exists) as the "most likely" value, that is, the value of

$\theta \in \Theta$ that maximizes the likelihood function $L(\theta; x_1, \ldots, x_n)$.

It is natural to estimate θ as the value of the parameter that maximizes $L(\theta)$. This leads to the *maximum likelihood estimate*, which may be formally defined as the value:

$$\hat{\theta} = \hat{\theta}(x) \tag{1}$$
$$\text{such that } L(\hat{\theta}) = \sup_{\theta \in \Theta} L(\theta).$$

Note that the symbol placed over a parameter, called a *caret* and denoted by $\hat{}$ (read as *hat*), is used to denote the maximum likelihood estimator (MLE), that is, $\hat{\theta}$. In practice, the MLE is the parameter value for which the observed sample is most likely.

In many circumstances, to find the maximum likelihood estimator $\hat{\theta}$, we need to solve an optimization problem using differential calculus. It is usually simpler to maximize the *log-likelihood* (ℓ) rather than $L(\theta)$,

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta).$$

since the logarithm is a monotonic transformation and therefore preserves the location of the maximum.

In simple cases, when dealing with a vector of parameters, i.e. $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_k)$ with $k > 1$, the maximization problem may be solved by finding the solution to the system of likelihood equations:

$$\frac{\partial \ell(\boldsymbol{\theta})}{\partial \theta_i} = 0, \quad i = 1, \cdots, k,$$

in this way, we obtain one or more candidates for the MLE, since the first-order condition (i.e., setting the first derivative equal to zero) is only a necessary condition for a maximum. We then check that the matrix of second derivatives, with elements,

$$\left. \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right|_{\theta = \widehat{\theta}} < 0, \quad i = 1, \cdots, k, \text{ and } j = 1, \cdots, k,$$

is negative definite[5].

Example: Consider a random variable X representing a success event, such as the event that the return of a bond in a given portfolio is greater than or equal to 3%. What is the proportion of bonds with returns higher than 3%? Suppose we have a random sample in which we observe

$$x_1 = 1, \quad x_2 = 0, \quad x_3 = 1.$$

After observing the realized values of the three random variables X_1, X_2, and X_3, we assume that they are distributed Bernoulli random variables with success probability $P(X \geq 3) = p$. We then seek the value of p that maximizes the following likelihood:

$$L(x_1 = 1, x_2 = 0, x_3 = 1; p) = P(X_1 = 1)P(X_2 = 0)P(X_3 = 1) = p^2(1-p).$$

Note that the above expression derives from the fact that $P(X_i = 1) = p^1(1-p)^{1-1}$.

$$L(\hat{p}) = \sup(L(p)) \quad p \in [0, 1].$$

The log-likelihood is

$$\ell(p) = \log(L(p)) = 2\log(p) + \log(1-p).$$

The first derivative is

$$\frac{\partial \ell(p)}{\partial p} = \frac{2}{p} - \frac{1}{1-p} = 0,$$

resulting in

$$\hat{p}_{ML} = \frac{2}{3}.$$

The second derivative is always negative; therefore, the maximum likelihood

[5] An $n \times n$ symmetric matrix A, is positive (negative) definite, iff $\mathbf{x}^T A\mathbf{x}$ is positive (negative) for any $\mathbf{x} \neq \mathbf{0} \in \Re^n$. One way to check whether a matrix is positive (negative) definite, is to check its eigenvalues: If all of the eigenvalues of a matrix are positive (negative), the matrix is positive (negative) definite.

estimates of this probability is 0.667, which corresponds to the expected proportion of bonds in the population with returns above 3%

Generally

$$\hat{p}_{ML} = \frac{\sum_{i=1}^{n} X_i}{n},$$

is the maximum likelihood estimator of the parameter p disposing of a random sample $X_1, \ldots, X_n$ drawn from a Bernoulli distribution.

The estimators obtained using the ML method have asymptotic (i.e., large n) properties:

- the MLEs can be biased (this does not apply to parameters estimated in the multiple linear regression model). However, the bias decreases as the sample size increases: they are asymptotically unbiased, meaning that as n grows, any bias they have approaches zero;
- the exact sampling distribution is sometimes difficult to obtain, but, as we shall see, a remarkable property is that their asymptotic distribution is generally Gaussian;
- the MLEs are consistent: as n increases, the estimator converges to the true parameter value;
- MLEs are asymptotically efficient: no other consistent estimators has a smaller asymptotic variance, and they to be closer to the true parameter value as the sample size increases.

Recall that the asymptotic properties of an estimator are valid when the sample size is large. MLEs belong to the class of *best asymptotically normal* estimators. The asymptotic sample distribution of the MLE, $\hat{\theta}_n$, under fairly general conditions, converges to a Gaussian distribution, so that

$$\frac{\hat{\theta}_n - \theta}{\sqrt{var(\hat{\theta})}} \rightarrow_d Z \sim N(0,1).$$

Here we introduce the Fisher information, which is used to quantify the uncertainty of model parameters through their standard errors. It is defined as

$$\bar{\mathrm{I}}(\theta) = \mathrm{E}_\theta\left[\left(\frac{\partial}{\partial\theta}\log f(x;\theta)\right)^2\right] = -\mathrm{E}_\theta\left[\left(\frac{\partial^2}{\partial\theta^2}\log f(x;\theta)\right)\right].$$

Here, the usual regularity conditions are assumed. For a sample of n independent observations from the assumed model, the overall Fisher information simply becomes $\mathrm{I}(\theta) = n\bar{\mathrm{I}}(\theta)$. This is a $p \times p$ matrix of the second derivatives of the log-likelihood function, with a change of sign. A large value of the Fisher information indicates that, in the parameter space, there is a region where the log-likelihood function attains very high values.

In general, $\bar{\mathrm{I}}(\theta)$ measures the amount of information about θ provided by a single observation, while $\mathrm{I}(\theta)$ represents the information provided by a sample of size n.

In this context, classic examples arise in the estimation of the mean of a Gaussian distribution (with known variance). For this model, we obtain the following results:

$$X \sim \mathrm{N}(\mu, \sigma^2) \qquad \bar{\mathrm{I}}(\mu) = 1/\sigma^2,$$

that is the Fisher information is equal to the reciprocal of the population variance (also called precision).

Fisher showed that the approximately normal distribution of the ML estimator is

$$\hat{\theta} \sim N\left(\theta, \frac{1}{\mathrm{I}(\theta)}\right).$$

The variance decreases as n increases, and thus the information increase.

In summary, the log-likelihood function is important not only for identifying the value of $\hat{\theta}$ that maximizes it, but also for using its curvature to determine the precision of $\hat{\theta}$ as an estimator of θ. MLE is the parameter value for which the data have the highest likelihood. A statistical method is said to satisfy the *likelihood principle* if it is based

solely on the likelihood function, reflecting how the data (and the design used to collect them) provide evidence about a parameter θ.

3.3 Bayesian methods

The Bayesian view of probability is that it represents a degree of belief about the event in question. An extensive introduction to Bayesian thinking and Bayesian data analysis can be found in Gelman et al. (2013), which covers a wide range of applications, with an emphasis on statistical modelling and simulation. The Bayesian approach assumes a prior distribution for the parameters, reflecting the information available about them before observing new data. This information may be based on previous studies or may reflect subjective prior beliefs, such as "experts" opinions. Alternatively, the prior distribution may be uninformative, so that inferential results are driven primarily by the observed data. In this framework, uncertainty about the parameters is expressed through their probability distributions.

The prior distribution is combined with the information provided by the data through the likelihood function to produce a posterior distribution for the parameters. Bayesian statistical inference is based on the posterior distribution, which represents the updated beliefs about the parameters given the observed data.

For a parameter θ, let $p(\theta)$ denote the prior distribution. In this context, the parameter is treated as a random variable. Let $f(\boldsymbol{y}|\theta)$ denote the likelihood function for the data, where the observations are $\boldsymbol{y} = (y_1, \ldots, y_n)$, given the parameter value. Let $g(\theta|\boldsymbol{y})$ denote the posterior distribution of θ, given the data. By Bayes' rule, the posterior probability is given by

$$g(\theta|\boldsymbol{y}) = \frac{f(\boldsymbol{y}|\theta)p(\theta)}{f(\boldsymbol{y})},$$

where $f(\boldsymbol{y}) = \int_\theta f(\boldsymbol{y}|\theta)p(\theta)d\theta$ is the marginal distribution of the data, obtained by integrating out the parameter. As a function of θ, $g(\theta|\boldsymbol{y})$ is proportional to $f(\boldsymbol{y}|\theta)p(\theta)$. Since this product determines the posterior distribution, we can often avoid computing $f(\boldsymbol{y})$ explicitly, which may be computationally difficult.

Once the data are observed, the posterior distribution of θ is determined by the

product of the likelihood function $f(\boldsymbol{y}|\theta)$ and the prior distribution of θ. When the prior distribution is is relatively flat (i.e., a non-informative prior), as is often assumed in practice, $g(\theta|\boldsymbol{y})$ has a shape similar to that of the likelihood function.

Bayesian inference based on the posterior distribution is parallel to classical inferences. For example, analogous to the classical 95% confidence interval for θ is an interval that contains 95% of the posterior density $g(\theta|\boldsymbol{y})$. This interval is called a posterior interval or credible interval. A simple posterior interval is obtained using the percentiles of $g(\theta|\boldsymbol{y})$: a 95% posterior interval for θ is the region between its 2.5 and 97.5 percentiles. The most common Bayesian estimator of θ is the posterior mean $\mathbb{E}(\theta|\boldsymbol{y})$. Except in a few simple cases, the posterior distribution cannot be computed analytically. In such situations, we can obtain an approximation using Markov chain Monte Carlo methods (Metropolis et al., 1953; Hastings, 1970). This approach relies on the fact that the posterior distribution can be approximated by a Markov chain whose stationary distribution is the posterior of interest. By iteratively simulating values from this Markov chain, we generate samples that are dependent but converge in distribution to the target posterior.

3.4 Inference based on resampling (bootstrap) method

Bootstrap is introduced as a basic tool for inference, as it can be used to assess estimation accuracy of any estimator (and algorithm), no matter how complicated. Modern computation power makes it possible to numerically approximate an infinite sequence of repeated samples. This is a remarkable feature of statistical inference in the computer age.

In statistics, a measure of accuracy of an estimate is provided by the associated standard error. It refers to the spread of a sampling distribution: the standard deviation of the sampling distribution of $\bar{X}$ is called standard error of $\bar{X}$. It describes how much $\bar{x}$ tends to vary from sample to sample of size n. The standard error is to distinguished from the population standard deviation, denoted as σ.

In machine learning bootstrap is employed to estimate extra-sample prediction error.

Many estimators (and algorithms) do not have closed-form formulas for computing standard errors. In 1949, the jackknife method was proposed, and later, in 1979, the non-parametric bootstrap was introduced. The advantage of the bootstrap over maximum likelihood-based approaches is that it allows us to estimate standard errors and other quantities even in settings where no analytical formulas are available.

In the non-parametric context, the only information available is provided by the sample, and the goal is to *"let the data speak as much as possible"*, extracting from them all the information about the parameter θ. In particular, resampling methods are based on the following idea: in the absence of prior information allowing us to make assumptions about $F(x)$, the data are extensively exploited through the iterative reuse of the sample.

The term *resampling methods* encompasses a range of modern techniques aimed at evaluating and/or improving the *accuracy* of an estimator, typically a complex one. For instance, such methods complex statistics are often employed for which it is not possible to derive an analytical expression for the standard error. The evaluation of accuracy is a fundamental step in the inference process: once an estimate $\hat{\theta}$ of the parameter ϑ has been obtained from the sample data, in accordance with statistical principles, it is necessary, in accordance with statistical principals, to assess its accuracy by evaluating some properties of the corresponding estimator $\hat{T}$. The bootstrap is a computational resampling method that treats the empirical distribution of the sample as if it were the population distribution. It is the best-known modern computer-based statistical method, introduced by Brad Efron. Efron was awarded the International Prize in Statistics in 2018[6]. The motivation pronounced by Sir David Cox is as follows: *"Because the bootstrap is easy for a computer to calculate and is applicable in an exceptionally wide range of situations, the method has found use in many fields of science, technology, medicine and public affairs"*. On that occasion, Prof. Efron also shared the following anecdote: *"I remember when going into statistics that the first year I thought that this will be pretty easy since I've dealt with math and that's supposed to be hard. But statistics was much harder for me at the beginning than any other field. It took years before I felt really comfortable."*

[6]See `https://www.eurekalert.org/pub_releases/2018-11/asa-ipi110618.php`

According to Davison and Hinkley (1997), the publication of Efron's first article on the bootstrap method (which was initially rejected by the journal) represented one of the central events in statistics, both because it synthesized several earlier ideas on sampling and because it laid the foundations for a new area of statistical analysis based on simulation methods. The name "bootstrap", assigned to the method by Efron himself, was inspired by the character Baron Munchausen from a popular tale, in which the protagonist falls into a lake and, unable to swim, attempts to pull himself out of the water "by pulling on the straps of his own boots".

This method has developed thanks to its simplicity, combined with the increasing availability of inexpensive computing power. In recent years, it has experienced rapid and widespread development, and it is now applied across various areas of statistics, including statistical inference, regression models, time series analysis, sampling theory, and many others.

Bootstrap method as originally proposed is as follows:

- A random variable X represents the characteristic on which we intend to make inference,
- Its cumulative distribution function $F(x)$ is completely unknown,
- The parameter of interest object of inference is θ, and we assume that it can be expressed as a function of the unknown F.

For example, $\theta = \text{Median}(X) = x_{0.5}$ is a function of F, since $x_{0.5} = F^{-1}(0.5) = \inf\{x|F(x) \geq 0.5\}$; $\theta = \mathbb{E}(X)$ is a function of F, as $\mathbb{E}[X] = \int_{\mathbb{R}} x\varphi(x)dx = \int_{\mathbb{R}} xdF(x)$, where $\varphi(x)$ denotes the probability density function or probability mass function of X.

The logic on which Bootstrap is based is the fusion of two techniques:

- the principle of plug-in,
- the Monte Carlo approximation, which involves the use of computer simulations.

At the base of the plug-in principle is the notion of the empirical cumulative distribution function $\hat{F}_n$. According to the plug-in principle, an estimate of $\vartheta = \vartheta(F)$ is

constructed by substituting (or *plugging in*) the empirical distribution function, that is, $\hat{\vartheta} = \vartheta(\hat{F}n)$. The sample mean, whose formal properties as an estimator of $\mu = \mathbb{E}(X)$ are well known, provides a simple example of the application of the plug-in principle.

As previously stated, the bootstrap can be viewed as an application of the plug-in principle when the goal is to evaluate the accuracy of an estimator for ϑ. One way to apply the plug-in principle in this context is to artificially reproduce the variability of $\hat{T}$ with respect to $\hat{F}_n$. In particular, since the variability with respect to F arises from sampling process used to generate the original data, it follows naturally that the variability with respect to $\hat{F}_n$ can be reproduced by drawing a *new* sample from the observed data, that is, by implementing resampling; see also, (James et al., 2013).

This is the idea underlying the bootstrap method and the reason why it is generally classified as a resampling method. The resampling procedure implemented by bootstrap is identical to that used for the original sample, or, more precisely, it mimics the original sampling process. In summary:

- Let $x_1, \ldots, x_n$ be a Bernoulli sample of fixed size n from X, on which inference about ϑ is to made;

- Let $x_1^*, \ldots, x_n^*$ denote bootstrap sample, that is, a sample of the same nature as the original one– Bernoulli and of size n – but drawn with replacement from the observed data $x_1, \ldots, x_n$. This sample $x_1^*, \ldots, x_n^*$ is called the bootstrap sample and can be viewed as a *randomized version* of the original sample, possibly including repeated observations.

- Let $\hat{\vartheta}^*$ be an estimate defined in the same way as $\hat{\vartheta}$ – that is, having the same functional form – but computed on the bootstrap sample. The quantity $\hat{\vartheta}^*$ is called a bootstrap replication of $\hat{\vartheta}$, or simply a replication.

The Bootstrap procedure for estimating $V_F(\hat{T})$ can be summarized as follows:

1. Starting from the original sample $x_1, \ldots, x_n$, you draw B bootstrap samples. Each bootstrap sample is obtained by sampling with replacement from the original

data, independently of the other bootstrap samples. Each bootstrap sample has size n, same as the original sample.

2. For each of the bootstrap sample, compute the statistic of interest $\hat{\theta}$, (e.g., mean, median, regression coefficient);

3. the replication $\hat{\vartheta}^*$ of $\hat{\vartheta}$ is calculated, obtaining the set of B values $\hat{\vartheta}_1^*, \ldots, \hat{\vartheta}_b^*, \ldots, \hat{\vartheta}_B^*$.

4. The bootstrap estimate of the variance of $\hat{\theta}$ is:

$$\hat{v}boot(\hat{T}) = \frac{1}{B-1} \sum_{b=1}^{B} (\hat{\vartheta}_b^* - \bar{\hat{\vartheta}}^*)^2, \tag{2}$$

where $\bar{\hat{\vartheta}}^* = \sum_{b=1}^{B} \hat{\vartheta}_b^*/B$ indicates the arithmetic mean of the B replications. Equation (2) defines the Bootstrap estimate for $V_F(\hat{T})$. The standard error is calculated as

$$\hat{s}eboot(\hat{T}) = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} (\hat{\vartheta}_b^* - \bar{\hat{\vartheta}}^*)^2},$$

which is assumed to be an estimate of the variability associated with the quantity of interest calculated in the original sample.

3.4.1 Confidence intervals: Percentile Method

The aim of confidence interval estimation is to find an interval of plausible values for unknown parameter θ based on sampling information. This can be achieved using a *confidence interval*, which specifies, instead of a single value for the parameter of interest, a **range of possible values** within which the parameter is estimated to lie with a certain level of confidence.

The probability that the method used to construct the interval produces an interval that contains the true value of θ is called the confidence level. This is a number chosen to be close to 1, such as 0.95 or 0.99. Once the data $Y = y$ are observed, the interval $[T_L(y), T_U(y)]$ is called a 100(1 - α)% confidence interval for θ, with lower and upper confidence limits $T_L(y)$ and $T_U(y)$. The probability $1 - \alpha$, called the confidence level,

is usually chosen close to 1, such as 0.90, 0.95 or 0.99. The corresponding probability α is the error probability. A probability such as 0.95 applies to the random interval $[T_L(Y), T_U(Y)]$, whose endpoints that are random variables, before the data are observed. *Probabilities apply to random variables, not to parameter values*.

Since the bootstrap distribution of an estimator $\hat{T}$ is a simulation, obtained by resampling, of the *true* distribution of $\hat{T}$, the bootstrap histogram can be considered a Monte Carlo approximation of the latter. The graphical representation of the bootstrap distribution via a histogram is based on the estimates obtained from the B replications, grouped into an arbitrary number of classes of arbitrary width. The quantiles of the bootstrap distribution can be interpreted as Monte Carlo approximations of the corresponding quantiles of the *true* distribution of the parameter θ.

The percentile method was proposed by Efron (Efron and Tibshirani, 1994) as one of the simplest methods for constructing confidence intervals, and it is based on the histogram of the bootstrap replications. Having fixed a confidence level $(1-\alpha)$, let $\hat{\vartheta}^*_{1-\alpha/2}$ and $\hat{\vartheta}^*_{\alpha/2}$ denote the quantiles of orders $1-\alpha/2$ and $\alpha/2$, respectively, of the bootstrap distribution. These quantiles can be interpreted as Monte Carlo approximations of the corresponding exact quantiles of the distribution of $\hat{T}$.

The interval defined by the following bootstrap quantiles

$$\left[\hat{\vartheta}^*_{\alpha/2}\,,\ \hat{\vartheta}^*_{1-\alpha/2}\right], \tag{3}$$

is the percentile bootstrap confidence interval for ϑ, with a confidence level (approximately) equal to the prefixed $(1-\alpha)$. For example, a 95% confidence interval obtained from 1000 bootstrap samples is given by the interval between the 25th and 975th ordered values of the bootstrap estimates.

3.5 Summary

- In statistics, a measure of accuracy of an estimate is provided by its associated standard error. In classical inference, this refers to the spread of the sampling distribution;
- When F is unknown, the bootstrap substitutes the empirical cumulative distribution function $\hat{F}$ for F;
- the bootstrap is automatic and perturbs the original data, producing deviations of $\boldsymbol{x}^*$ from $\boldsymbol{x}$;
- A number of bootstrap replicates $B = 200$ is generally sufficient for standard errors estimation. Larger values, such as $B = 2000$, are typically required for bootstrap confidence intervals;
- For large samples, maximum likelihood theory yields:

$$\hat{\theta} \sim N\left(\theta, \frac{1}{nI(\theta)}\right),$$

  and in practice this is implemented by estimating standard error via

$$\hat{se}(\hat{\theta}) = (nI(\hat{\theta})^{-1/2})$$

  ;
- Note that the number of distinctive bootstrap resamples is

$$\binom{2n-1}{n}$$

  so that, for example, if $n = 10$, there are 92,378 possible distinct samples;
- Confidence intervals are often constructed to achieve approximately 95% coverage as

$$\hat{\theta} \pm 1.96\hat{se}(\hat{\theta}),$$

but these intervals may be inaccurate because they are symmetric about $\hat{\theta}$. Using the bootstrap distribution of an estimator $\hat{\theta}$, we can improve upon standard intervals by accounting for the shape of the distribution. In particular, percentile confidence limits provide more accurate inference; see (Efron and Hastie, 2021, Chapter 11).

3.6 Jackknife

The history of resampling methods began in 1949 with the first jackknife proposal by M. H. Quenouille (Quenouille et al., 1949), introduced to evaluate the accuracy of a complex estimator. Although it is based on samples extracted from the original data, early resampling methods were designed to require minimal computation, since modern digital computers were not yet available. In particular, the jackknife required relatively little processing.

Let $x_1, \ldots, x_n$ be a Bernoulli sample size n from a random variable X with cumulative distribution function $F(x; \vartheta)$ and let $\hat{\vartheta}$ be an estimator of ϑ. In its original formulation, the jackknife considers n subsamples of size (n-1) obtained from the original sample by systematically deleting the $i = 1$-th observation x_i, for $i = 1, \ldots, n$.

These subsamples, called Jackknife samples have the generic form

$$x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n.$$

The jackknife method (algorithm) consists of the following steps. For each i-th jackknife sample, the quantity $\hat{\vartheta}(i)$ is computed in the same way as $\hat{\vartheta}$, with the only difference being that the sample size is $(n-1)$ instead of n. The procedure is repeated n times, once for each jackknife sample, yielding n estimates $\hat{\vartheta}(i)$, $i = 1, \ldots, n$.

4 Multiple linear regression

Statistical methods for multiple variables typically analyze how an outcome or response variable, is associated with, or can be predicted by, the values of a set of *explanatory variables*, also referred to as *predictors, covariates, independent variables, exogenous variables, or terms*. Regression models specify how these two set of variables are related under certain assumptions. They can be estimated for explanatory and/or predictive purposes.

In the following, we consider a continuous response variable and present the key ideas of the multiple linear regression, a method developed in the pre-computer era of statistics that is widely used due to its simplicity and interpretability in describing how one or more outputs are associated with inputs. Although more complex non-linear models are available, multiple linear regression often performs competitively for predictive purposes. Moreover, if has been extended in several directions since its original formulation; see, among others, Stasinopoulos et al. (2024). This approach is classified as a supervised learning method, since the presence of the outcome variable guides the learning process, and predictions of the outcome variable guides the learning process, and predictions are made on the basis of observed features that are temporally or logically related to the response variable. In the linear multiple regression model, we relate a quantitative random variable to a linear combination of quantitative and categorical variables. The model allows us to estimate the conditional effect of each explanatory variable on the response. Note that the covariates may be continuous (such as age), spatial (e.g., coordinates representing geolocation), or categorical variables of different types.

An example in which the multiple linear regression model can be applied is the following. The yield of wheat per acre for the month of April is thought to be related to the rainfall. A researcher randomly select acres of wheat and records temperature, the amount of rainfall, and the bushels of wheat per acre. In this example, the *response variable* is the yield of wheat, while the *explanatory variables* are temperature and rainfall. The mean yield per acre is assumed to be linearly related to both variables. Field yields are assumed independent, and their standard deviation is approximately constant

across different levels of temperature and rainfall.

In the following, we consider the *conditional probability distribution* describing the variation of the response variable Y around its conditional expectation for each value of the covariates x_1 and x_2. We treat Y as a random variable observed at different values of x_1 and x_2. We treat Y as a random variable observed at different values of x_1 and x_2; accordingly, we use uppercase notation for the response and lowercase notation for the covariates. Let (x_{i1}, x_{i2}, Y_i) denote the values of x_1 and x_2, and Y for observation i, $i = 1, \ldots, n$. We also define the vector $\boldsymbol{y} = (y_1, y_2, \ldots, y_n)^\intercal$, where $^\intercal$ denotes the transpose, so that $\boldsymbol{y}$ is a $n \times 1$ column vector.

First, we introduce the linear regression model along with the underlying assumptions in the case of two explanatory variables. The exposition is then extended to the case of multiple explanatory variables. Some of the inferential and interpretative aspects of the linear model are addressed through parameter estimation methods. Model validation involves checking the underlying assumptions, as well as assessing the accuracy of the estimates and predictions.

4.1 Model specification

The following topics are discussed:

- model assumptions
- estimation of the model parameters
- residuals analysis
- confidence intervals and hypothesis testing for the regression coefficients
- prediction of values.

In the multiple linear regression model, the expected value of the response variable is specified conditional on the observed values of the explanatory variables. In the following discussion, it is assumed that Y is absolutely continuous. The following

functional form for the response variable, referring to the i-th unit, is assumed

$$Y_i = f(x_{i1}, x_{i2}) + \epsilon_i, \quad i = 1, \ldots, n \tag{4}$$

where f is an unknown function, and x_{i1} and x_{i2} are the covariates (or feature measurements) of the i-th unit. The term ϵ_i is referred to as the random component or unit-specific error; it is a random variable that enters the model in an additive manner. Each observation has its own ϵ_i, whose sign reflects whether the observation lies above or below the conditional expected value. This term, also called a disturbance, quantifies the *deviation between the structural part of the model and the observed response variable*.

In the basic formulation, the error term is assumed to be a continuous random variable with the following moments:

$$\mathbb{E}[\epsilon_i] = 0, \qquad \mathrm{Var}(\epsilon_i) = \sigma^2,$$

where the variance $\sigma^2 > 0$ is constant across units. The errors are assumed to be independent of each other and, consequently, uncorrelated; therefore, their covariance are zero.

If we specify the function as in the Equation (4) as linear and additive, the model becomes:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i.$$

The regression coefficients collected in the vector $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^\intercal$ are the model parameters:

- β_0 represents the intercept,
- β_1 and β_2 are the regression coefficients referred to the two explanatory variables.

An alternative model formula is the following

$$\mathbb{E}[Y] = \mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

Note that the model is linear, and the covariates can come from different sources: continuous variables, transformations of these variables (e.g., logarithm or square-root), basis expansions, dummy variables, or interactions between variables.

4.1.1 Matrix notation

We introduce the matrix notation for the multiple linear regression model, illustrating the formulation for a model with three explanatory variables and$n > 3$ observations. The model can be represented by the following set of equations:

$$\begin{aligned} Y_1 = &\quad \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13} + \epsilon_1 \\ Y_2 = &\quad \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \beta_3 x_{23} + \epsilon_2 \\ \vdots &\qquad\qquad \vdots \\ Y_n = &\quad \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \beta_3 x_{n3} + \epsilon_n. \end{aligned}$$

In terms of matrix algebra the system of equations is

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_i \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{i1} & x_{i2} & x_{i3} \\ \vdots & \vdots & \vdots & \\ 1 & x_{n1} & x_{n2} & x_{n3} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_i \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model is expressed as

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where

- $\boldsymbol{Y}$ is the vector of dimension $n \times 1$ collecting the response variables;

- $\boldsymbol{X}$ is the matrix of explanatory variables of dimensions $n \times (p+1)$ and it is a non-stochastic matrix of full rank (i.e. the explanatory variables are linearly independent);

- $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)'$ is the vector of the model parameters of dimension $(p+1) \times 1$;

- $\boldsymbol{\epsilon}$ is the vector of the error terms where each ϵ_i is referred to the i-th observation with $i = 1, \ldots, n$.

  Model assumptions require that:

  (i) each error term is a random variable with zero mean, $\mathbb{E}[\epsilon_i] = 0$, and constant variance $\mathrm{Var}(\epsilon_i) = \sigma^2$, $\forall i$;

  (ii) the errors are independent, so that $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$, i.e., $\mathrm{Var}(\boldsymbol{\epsilon}) = \sigma^2 \boldsymbol{I}_n$ (homoscedasticity), where $\boldsymbol{I}_n$ denotes the $n \times n$ identity matrix.

4.2 Model specification with two explanatory variables

We consider the data (x_i, y_i), for $i = 1, \ldots, n$, and a *prediction equation* of the following type

$$\hat{\mu}_i = \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2}. \tag{5}$$

This equation provides an estimates of the conditional expected value of the response $\mathbb{E}[Y] = \mu_i$:

- where $\hat{\mu}_i = \hat{y}_i$ is called fitted value for unit i;

- the realized value of $\hat{\beta}_0$ is the estimate of an (unknown) parameter of the model called intercept;

- the realized values of $\hat{\beta}_1, \hat{\beta}_2$ are the estimates of the (unknown) parameters of the model: β_1 represents the conditional effect of the explanatory variable x_1 and β_2 represents the effect of x_2. These parameters are called partial regression

coefficients, since each measures the change in the expected value of the response for a one-unit increase in x_j $(j = 1, 2)$ while controlling for (or adjusted for) the other variables included in the model.

Let z_i be such that

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + z_i,$$

$$z_i = y_i - \hat{y}_i,$$

that is z_i called residual referred to observation i.

4.3 Least squares estimation method

The least squares method, together with the maximum likelihood method, is the mathematical procedure for deriving parameter estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ by minimizing the following Euclidean distance

$$D(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

The least-squares estimates of the model parameters are the values that minimize the sum of the squared deviations between the observed and fitted values, thus providing a regression plane that is maximally consistent with the observed points with respect to the chosen Euclidean distance. This method is appropriate when the observations represent independent random draws from the population.

The system of equations becomes

$$D(\beta_0, \beta_1, \beta_2) = \text{argmin} \sum_{i=1}^{n} [y_i - (\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})]^2,$$

which is a system of $p+1$ partial derivatives. The values of the parameters that solve the system are denoted with $(\hat{\beta}_1, \hat{\beta}_2)$. These values minimize the least squares criterion and are referred to as the ordinary least squares (OLS) estimators of the model parameters.

The parameter $\hat{\beta}_0$ is obtained as

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2.$$

Note that the plane certainly passes through the point $(\bar{y}, \bar{x}_1, \bar{x}_2)$ which is defined as the *center of gravity* of the data.

Figure 13 illustrates, as an example, the observed data (the points) and the linear regression plane fitted using least-squares estimation (represented by the values on the blue grid).

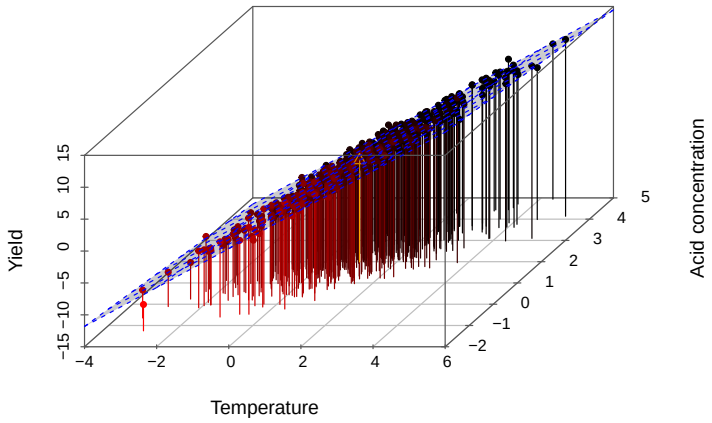


Figure 13: *Example of observed data and a fitted regression plane. The data relate to the yield of a chemical process as a function of temperatures and acid concentration. The average values of the there variables are highlighted in orange.*

We notice that:

- $\hat{\beta}_0$ is the estimator of the intercept and provides the conditional expected value of the response variable when the two explanatory variables are set equal to zero ($x_1 = x_2 = 0$);

- $\hat{\beta}_1$ is the estimator of the conditional regression coefficient referred to x_1. It measures the influence of this variable on the response holding fixed the other

variable x_2. It indicates the expected change in the response variable according to the model when x_1 increases by one unit while holding fixed the value (or operating at parity of, while controlling for other variable in the model) x_2; this is a conditional effect, adjusted for the other explanatory variables.

- $\hat{\beta}_2$ is the estimator of the conditional regression coefficient referred to x_2. The interpretation is similar to the previous one.

Note that from

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2,$$

it follows that if x_1 increases by 1 unit while x_2 remains constant we have

$$\hat{y}' = \beta_0 + \beta_1 (x_1 + 1) + \beta_2 x_2,$$

then the variation can be written in terms of the parameter β_1 as follows

$$(\hat{y}'_1 - \hat{y}) = \beta_1 (x_1 + 1 - x_1) = \beta_1.$$

The same expression is obtained for β_2 once we consider a unit change of x_2 with x_1 left constant.

To **summarize** the linear regression model:

- the associations between a continuous response variable and two or more covariates is assumed to be linear;
- the functional form linking the response to the explanatory variables is linear and additive and it includes an error term;

- Inference is made by conditioning on the observed values of the explanatory variables.
- The model is defined for the reference population and explains the variability of the response according to the covariates.
- The errors are random variables with constant variance and are uncorrelated with each other.
- The variance-covariance matrix of the explanatory variables must be of full rank; that is, the explanatory variables must not be linearly dependent.
- The parameters are estimated using the method of least squares or maximum likelihood.
- Applying the principle of least squares yields the system of normal equations, whose solutions provide the estimates of the model parameters.
- The least-squares estimates provide the prediction equations that are closest to the data, minimizing the sum of squared residuals.

In the following, after describing the residuals and their properties, we will address several issues related to model evaluation while preserving its simplicity: (i) variable selection; (ii) detection of unusual and influential observations; (iii) inference on the model parameters; (iv) prediction.

Residuals are used as a diagnostic tool to assess whether the main features of the data are adequately captured by such a simple linear model.

4.4 Residuals

For each observation i, we define the residual as difference between the observed value and the fitted value

$$z_i = y_i - \hat{y}_i.$$

Residuals are examined empirically to assess the validity of the model assumptions.

The following three properties of the residuals derive from the method of leasts squares used to estimate the model parameters:

- I Property: the sum of the residuals is zero $\sum_{i=1}^{n}(z_i) = 0$, when the model includes the intercept. That is, the sum of fitted values coincides with the sum of the observed values.

- II Property: the residuals are orthogonal to each explanatory variable, that is, $\sum_{i=1}^{n}(z_i x_{i1}) = 0$ and $\sum_{i=1}^{n}(z_i x_{i2}) = 0$. Therefore, the underlying random variable Z is uncorrelated with x_1 and x_2.

- III Property: z and $\hat{y}$ are uncorrelated, that is,$cov(z, \hat{y}) = 0$. Therefore residuals are orthogonal to the fitted values.

4.5 Decomposing variability: Analysis of variance

The variance of the error terms, assumed to be constant and denoted by σ^2, is an unknown parameter referred to as the error variance. In a linear model having an intercept and p explanatory variables, an *unbiased* estimator of the error variance is given by the following quantity

$$S^2 = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n - (p+1)}.$$

It can be shown that $\mathbb{E}[S^2] = \sigma^2$.

A commonly used index for evaluating the goodness of fit of a multiple linear regression model is based on the *residual standard error* (RSE), which is defined as

$$S = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n - (p+1)}}.$$

A lower value of S indicates a better model, as it suggests that the observed and fitted values are in close agreement.

Due to fundamental properties of the least squares estimators, it is possible to derive the proportion of the total variance that is explained by the multiple linear regression model, thereby providing a measure of goodness-of-fit. First, we consider the quantity defined as the total sum of squares (TSS), which is measured by the empirical variance of the observed responses, that is

$$TSS = \sum_{i=1}^{n}(y_i - \bar{y})^2,$$

It yields the following decomposition:

- adding and removing the fitted values results as

$$\sum_{i=1}^{n}[(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2,$$

- rewriting

$$\sum_{i=1}^{n}\Big[(y_i - \hat{y}_i)^2 + (\hat{y}_i - \bar{y})^2 + 2(y_i - \hat{y}_i)(\hat{y}_i - \bar{y})\Big],$$

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + 2\sum_{i=1}^{n}(y_i - \hat{y}_i)(\hat{y}_i - \bar{y}),$$

we obtain the residuals

$$\sum_{i=1}^{n}(z_i^2) + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 + 2\Big(\sum_{i=1}^{n} z_i\hat{y}_i - \bar{y}\sum_{i=1}^{n} z_i\Big),$$

and applying to the properties of residuals we have that $\sum_i z_i\hat{y}_i = 0$ (Property III) while $\sum_{i=1}^{n} z_i = 0$ (Property I). Therefore

$$\sum_{i=1}^{n}(z_i^2) + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2,$$

the total sum of squares decomposes into residual and explained sum of squares.

4.5.1 Multiple R-Squared

R-squared is the following

$$R^2 = \frac{TSS - SSE}{TSS} = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2 - \sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{SSE}{TSS},$$

where SSE is Sum of Squared Errors and TSS is the Total Sum of Squares. It is the proportion of explained variability of the response and it takes values between 0 and 1 in linear regression models with an intercept.

- $R^2 = 1$ when residual sum of squares is zero, that is observed and fitted values are equal;
- $R^2 = 0$ when explained variability of the model is null. The fitted plane reduces to the arithmetic mean $\hat{\beta}_0 = \hat{y} = \bar{y}$. The explanatory variables are irrelevant to explain the variability of the response;
- A value of 0.7 is interpreted as the model describing 70 per cent of the overall variability of the response.

The index measures the ability of the explanatory variables to account for the variability of the response. For example, a value of 0.7 indicates that the model explains 70% of the total variability in the response. A value close to 1, however, does not imply that the model is correctly specified. In the following, we perform diagnostic checks based on the residuals and assess the model assumptions to determine whether the model provides reliable results, regardless of the value of R^2.

The adjusted R^2 corrects for the tendency of R^2 to increase when additional explanatory variables are included in the model. Its expression is

$$\bar{R}^2 = 1 - \frac{n-1}{n-p-1}(1 - R^2),$$

where n is the sample size and p is the number of explanatory variables (excluding the intercept), with $n - p - 1$ degrees of freedom when there are $(p + 1)$ parameters in the

model. This index is typically slightly smaller than the ordinary R^2 and, unlike R^2, it does not increase monotonically when additional explanatory variables are included in the model. However, the penalty for adding a new covariate is often relatively small.

4.6 Diagnostics for unusual and influential observations

Inferential conclusions drawn from a multiple linear regression model are valid only after an adequate assessment of model fit. *However, the assumptions underlying the model should be satisfied at least approximately*.

In the classical linear model, it is necessary to assess the assumptions of homoscedasticity, independence (uncorrelated errors), and, possibly, normality of the errors, as well as the linearity of the relationship between the predictors and the response. The properties of residuals are used to detect potential model inadequacies. Graphical inspection of residuals helps to assess whether the estimated model is able to capture the underlying structure of the data and to identify observations that may be problematic. Note that there is no universally accepted definition of outliers. We define an outlier as an observation that has a substantial impact on estimation and inference when removed from the dataset.

We expect residuals not to exhibit systematic trends or specific patterns, as the presence of clusters or structured arrangements of points may indicate violations of the assumptions underlying the fitted model.

In particular, we consider the following residual plots:

- Representing units $i = 1, \ldots, n$ on the x-axis against the residuals (z_i) on the y-axis. In this figure we expect the values to be random over the plane. There must be a balance of positive and negative residuals. The variability of the residuals must appear constant across units. When residuals referring to adjacent units show up with particularly similar values, this is a sign of a possible violation of the assumption of no correlation among error terms. If, for example, one fits a linear model to the time series of daily oil prices, the residuals will undoubtedly exhibit a systematic pattern rather than the expected random behavior because

the prices are temporally correlated, and this linear regression model does not allow accounting for this dependency structure, as the data generating process changes over time.

- The plot of the fitted values ($\hat{y}_i$) on the x-axis against the residuals (z_i) on the y-axis. We expect that the points do not show systematic trends, so that is positive and negative residuals alternate randomly. If a clear U-shaped pattern is present, this indicates a possible violation of the assumption of linearity of the model. In order to better interpret extreme observations or *outliers* that may occur in this plot, a transformed version of the residual is considered, namely, standardized or studentized residuals.

- The plot of the residuals on the y-axis against each explanatory variable on the x-axis is used to reveal possible heteroskedasticity (errors terms with non-constance variance) and nonlinearity in the association between the covariate and the response variable. Moreover, if one or more values are particularly distant from the others in the plane, they are called leverage points. These observations can strongly influence the model fit, and the estimated parameter values may become unreliable. The consequences of ignoring heteroscedastic error variances are that the estimated standard errors of the regression coefficients may be biased (often underestimated or overestimated) and hypothesis tests and confidence intervals may be invalid, and the model will present high uncertainty.

  In particular, in the plots of the residuals mentioned above we should note:

  - When positive residuals are greater than negative ones: in this case the distribution of the random response variable from which data have been potentially generated may be skewed;
  - When residuals do not fluctuate randomly on the plane but follow a specific pattern as shown in Figure 14 where they decrease and increase: this is a sign of nonlinear associations between the response variable and the explanatory variables;

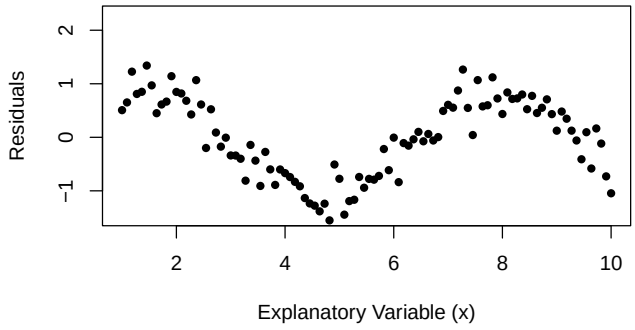


Figure 14: *Plot of the residuals when the assumed model is misspecified model (assuming a linear effect).*

  - When the variability of the residuals is not constant: the assumption of constant variance of the error terms (homoscedastity) might not be fulfilled. If the scatter plot of fitted values against residuals has a funnel shape (initially, the points show less variability, and then they show higher variability), this indicates possible heteroscedasticity in the model errors. Figure 15 shows residuals with increasing variance, indicating that this assumption is violated.

- The leverage is a measure of the potential influence of an observation on the fit. Observations for which the explanatory variables are far from their means have greater potential influence on the least squares estimates (Tukey, 1962, 1977). The following measure is considered

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}, \quad i = 1 \ldots, n.$$

  This diagnostic is nonnegative $0 \leq h_{ii} \leq 1$, and large leverage values indicate an unusual covariate value $\boldsymbol{x}_i$, that is the observation x_i is far from the average $\bar{x}$. Large leverage does not necessarily lead to problems, as a rule of thumb, observations with $h_{ii} > 2p/n$ (twice the average) should be examined more closely, see (Fahrmeir et al., 2022) for additional details.

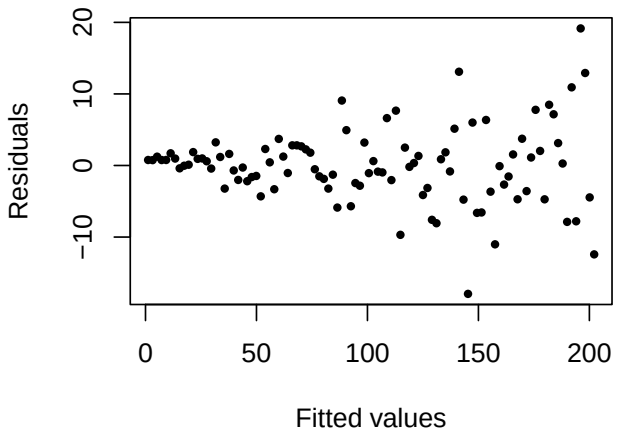


Figure 15: *Plot of the residuals when the assumption of constant variance of the errors is violated.*

- The point corresponding to a particularly high value of the standardized or studentized residual is considered a potential outlier, and the measure is

$$m_i = \frac{z_i}{s\sqrt{1-h_{ii}}},$$

where $h_{ii} = \mathbf{x}_i^\top (X^\top X)^{-1}\mathbf{x}_i$.

where s is an estimate of the conditional standard deviation σ. Outliers are observations that do not seem to follow the same data-generating process as the other observations, and they are detected by large residuals.

It is not always appropriate to eliminate observations corresponding to outliers automatically once they have been detected. For example, as reported by (Faraway, 2016, pg 89) NASA had launched a satellite called Nimbus 7[7] as early as 1978 in order to record variations in the composition of the atmosphere. The data provided by the satellite, however, failed to detect the hole in the ozone layer already present because the analyses identified outliers and removed them automatically. According to (Page 89 Faraway, 2016), the British monitoring system first detected

[7] https://en.wikipedia.org/wiki/Nimbus_7

the ozone hole in Antarctica in 1985, watch also the following video `https://www.youtube.com/watch?v=7Vicds7WqEc&t=10s`.

- An influent value contributes highly to determine the fitted values. A useful measure for determining these values is named Cook's distance. It measures the changes on the estimated coefficient $\hat{\beta}_j$ when a single observation i is removed from the data. For residual z_i and levarage h_{ii} the Cook's distance is

$$d_i = \frac{z_i^2 h_{ii}}{(p+1)s^2(1-h_{ii})^2}.$$

  It is nonnegative and offers a measure of standardised distance between fitted values obtained with all observations and that obtained without the i-th observation (leave one out as the Jacknife approach). Larger values of Cook's distance require both large standardized residuals and leverages. In general, if $d_i > 1$ the observation referring to the i-th unit may be an influential value and it should be examined.

5 Inference for the multiple linear regression model

We use matrix notation, assuming that the errors follow a multivariate Gaussian distribution:

$$\boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \sigma^2 \boldsymbol{I}_n).$$

Consequently, the conditional distribution of the response variable is also a multivariate Gaussian distribution:

$$\boldsymbol{Y}|\boldsymbol{X} \sim N_n(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}_n).$$

In the following, we provide the expressions for least squares method within the normal multiple linear regression model. The model matrix, as introduced in Section 4.1.1, is given by:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

Least squares estimates are obtained by minimizing the sum of squared residuals:

$$\operatorname{argmin} \sum_{i=1}^{n} (Y_i - \boldsymbol{X}_i \boldsymbol{\beta})^2 = \arg\min ||\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}||^2, \tag{6}$$

where the corresponding normal equations are

$$(\boldsymbol{X}'\boldsymbol{X})\boldsymbol{\beta} = \boldsymbol{X}'\boldsymbol{y}.$$

Since $(\boldsymbol{X}'\boldsymbol{X})$ is positive definite and invertible, the normal equations have a unique solution given by the ordinary least squares estimator:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y},$$

where the least squares estimate is a linear function of the observations for $\boldsymbol{y}$.

Premultiply on both sides of the equation by $\boldsymbol{X}$ yields the fitted values $\hat{\boldsymbol{y}}$

$$\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y},$$

where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is an $n \times n$ symmetric and idempotent matrix, named the hat matrix. It linearly transforms $\boldsymbol{Y}$ onto the space defined by $\boldsymbol{X}$. The following expression holds

$$\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y} = \boldsymbol{X}\hat{\boldsymbol{\beta}}.$$

Each linear model is associated with a unique projection matrix. Note that the diagonal element h_{ii} in row i and column i of the hat matrix $\boldsymbol{H}$ is the leverage of observation i.

Residuals can be also expressed in matrix notation as follows

$$\boldsymbol{z} = (\boldsymbol{y} - \hat{\boldsymbol{y}}) = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{y},$$

which satisfy the condition $\boldsymbol{X}'(\boldsymbol{y} - \hat{\boldsymbol{y}})$. Note that letting $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{H})$, where $\boldsymbol{M}$ is also a symmetric and idempotent matrix, the sum of squared residuals can be written as

$\boldsymbol{y}'\boldsymbol{M}\boldsymbol{y}$.

According to the previous expressions

$$\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{z} = \boldsymbol{X}\hat{\boldsymbol{\beta}} + \boldsymbol{z}.$$

5.1 Properties of the least squares estimators

One justification for the least squares estimators is that they coincide with the maximum likelihood estimators under the usual normality assumption. Hence, they are more efficient than estimators obtained using other criteria. Least squares therefore provide the best possible estimator of $\boldsymbol{\beta}$ within the class of linear unbiased estimators.

The corresponding likelihood function for σ^2 and $\boldsymbol{\beta}$ is given by

$$L(\boldsymbol{\beta}, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})\right),$$

and maximizing this likelihood is equivalent to minimizing the residual sum of squares, which leads to the normal equations.

The estimators satisfy the following properties:

- The least squares estimator is unbiased for $\boldsymbol{\beta}$, that is,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}.$$

Proof Since $\boldsymbol{X}$ is a non-stochastic matrix of full rank,

$$\begin{aligned} \mathbb{E}[\hat{\boldsymbol{\beta}}] &= \mathbb{E}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}], \\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathbb{E}[\boldsymbol{y}]. \end{aligned}$$

Moreover,

$$\begin{aligned} \mathbb{E}[\boldsymbol{y}] &= \mathbb{E}[\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}], \\ &= \boldsymbol{X}\boldsymbol{\beta} + \mathbb{E}[\boldsymbol{\epsilon}] = \boldsymbol{X}\boldsymbol{\beta}. \end{aligned}$$

Therefore,

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] \;=\; (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}.$$

For the Gauss–Markov theorem, among all linear unbiased estimators, the least squares estimator has minimum variance and is therefore called the best linear unbiased estimator (BLUE). The estimator $\hat{\boldsymbol{\beta}}$ is the best among all *linear and unbiased* estimators and is therefore the most efficient. The Gauss–Markov theorem is also used to obtain an optimal prediction for a future observation y_w with a given covariate vector $\boldsymbol{x}_w$.

- The variance-covariance matrix of the least squares estimator is given by

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}.$$

Proof Since $\boldsymbol{X}$ is a non-stochastic matrix of full rank,

$$\begin{aligned}\mathrm{var}[\hat{\boldsymbol{\beta}}] &= \mathrm{var}[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}],\\ &= (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\mathrm{var}(\boldsymbol{y})\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1},\end{aligned}$$

where $\mathrm{var}(\boldsymbol{y}) = \sigma^2\boldsymbol{I}_n$. Denoting $\boldsymbol{A} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$, we have

$$\begin{aligned}\mathrm{var}(\hat{\boldsymbol{\beta}}) &= \mathrm{var}(\boldsymbol{A}\boldsymbol{y}) = \boldsymbol{A}\,\mathrm{var}(\boldsymbol{y})\,\boldsymbol{A}' = \sigma^2\boldsymbol{A}\boldsymbol{A}'\\ &= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}.\end{aligned}$$

- Under the Gaussian distribution assumption for the error terms, the estimator $\hat{\boldsymbol{\beta}}$ also follows a multivariate Gaussian distribution. The distribution of the least

squares estimator is

$$\hat{\boldsymbol{\beta}} \sim N_{p+1}\left(\boldsymbol{\beta}, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}\right),$$

and each component $\hat{\beta}_k$ has a univariate Gaussian distribution,

$$\hat{\beta}_k \sim N(\beta_k, \sigma^2_{kk}),$$

where σ^2_{kk} is the k-th diagonal element of $\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}$, for $k = 1, \ldots, p+1$.

5.2 Inference on the model parameters

An unbiased estimator of σ^2 is given by

$$S^2 = \frac{SSE}{n-p-1} = \frac{(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})'(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})}{n-p-1} = \frac{\boldsymbol{z}'\boldsymbol{z}}{n-p-1},$$

where the numerator is the sum of squared errors and $n-p-1$ is the number of degrees of freedom; n is the sample size and p is the number of covariates (when $p = 2$, we are estimating a regression plane). For the normal linear model, it follows that

$$\frac{(n-p-1)S^2}{\sigma^2} \sim \chi^2_{n-p-1},$$

that is, the scaled estimator has a central chi-squared distribution with $n-p-1$ degrees of freedom.

Therefore, the distribution of S^2 is

$$S^2 \sim \frac{\sigma^2 \chi^2_{n-p-1}}{n-p-1}.$$

Due to the previous assumption on the error terms,

$$\boldsymbol{\epsilon} \sim N_p(\boldsymbol{0}, \sigma^2 \boldsymbol{I}),$$

where $\boldsymbol{0}$ is a vector of zeros and $\boldsymbol{I}$ is the $p \times p$ identity matrix, with ones on the main

diagonal and zeros elsewhere. Thus, the covariance matrix is $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}$.

The residuals $\boldsymbol{z}$ may be written in terms of the projection matrix $\boldsymbol{H}$ as

$$\boldsymbol{z} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y},$$

where, as introduced before,

$$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'.$$

Then,

$$\boldsymbol{z} \sim N_p(\boldsymbol{0}, \sigma^2(\boldsymbol{I} - \boldsymbol{H})),$$

since $\mathbb{E}(z_i) = 0$ and $\text{var}(z_i) = \sigma^2(1 - h_{ii})$, where h_{ii} is the i-th diagonal element of $\boldsymbol{H}$.

5.2.1 Testing that all effects are equal to zero: the F-test

The test statistic used to assess the global null hypothesis is called the F-test. The null hypothesis is

$$\beta_k = 0 \qquad \text{for all } k = 1, 2, \ldots, p.$$

If this hypothesis is not rejected, then each response value, for $i = 1, \ldots, n$, can be expressed simply as a signal (the intercept β_0) plus an error term (ϵ_i). Therefore, the response is independent of all the covariates, and this model is called the null model.

Consider the following multiple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i, \qquad i = 1, \ldots, n.$$

If $\beta_k = 0$, this implies that the response variable is conditionally independent of the corresponding covariate X_k, given all the other explanatory variables included in the model.

The null hypothesis is

$$H_0 : \beta_1 = \beta_2 = 0,$$

stating that all regression coefficients are zero except for the intercept. Obviously, if the explanatory variables have been chosen as potentially accounting for the observed variability in the response, one would expect to reject the null hypothesis.

The *full model* with p explanatory variables is

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

and under H_0 it is compared with the following *null model*:

$$\boldsymbol{Y} = \mathbf{1}\beta_0 + \boldsymbol{\epsilon},$$

where none of the explanatory variables has any effect on the response. Under the null hypothesis,

$$\boldsymbol{Y} \sim N_n(\beta_0 \mathbf{1}, \sigma^2 \boldsymbol{I}).$$

The observed sum of squared errors is

$$SSE = \sum_i (y_i - \hat{y}_i)^2,$$

while for the null model we have

$$TSS = \sum_i (y_i - \bar{y})^2.$$

Larger values of $(TSS - SSE)$ provide stronger evidence against H_0. This difference depends on the units of measurement of y, and its value tends to be larger when the error variance σ^2 is larger.

The decomposition

$$TSS = SSR + SSE = (TSS - SSE) + SSE$$

expresses the total sum of squares as the sum of two components, and the quantities

$$(TSS - SSE)/\sigma^2 \qquad \text{and} \qquad SSE/\sigma^2$$

are independent random variables whose distributions do not depend on the units of measurement.

Under the previous assumptions for the model with p explanatory variables, the distribution of the residual sum of squares satisfies

$$\frac{SSE}{\sigma^2} \sim \chi^2_{n-p-1}.$$

Under the null hypothesis,

$$\frac{TSS}{\sigma^2} \sim \chi^2_{n-1}.$$

To test H_0, the distribution of the test statistic is obtained from the ratio of two independent chi-squared variables divided by their respective degrees of freedom. In particular,

$$\frac{(TSS - SSE)/p}{SSE/(n-p-1)}$$

has, under H_0, a Fisher–Snedecor distribution with $df_1 = p$ and $df_2 = n - p - 1$ degrees of freedom. Therefore,

$$F = \frac{(TSS - SSE)/p}{SSE/[n-(p+1)]} \sim F_{p,n-p-1}.$$

Under H_0, larger values of $(TSS - SSE)$ produce larger values of the F statistic and therefore stronger evidence against H_0. Let α denote the significance level. We reject the null hypothesis if the test statistic is greater than the $(1-\alpha)$-quantile of the corresponding F distribution.

When the observed value of the test statistic, say f_{obs}, is greater than the critical value from the reference distribution $F_{p,n-p-1}$ at the chosen significance level, the null hypothesis is rejected. Of course, rejection of the null model does not imply that the full model has been correctly specified.

Software for fitting normal linear models reports results in terms of the p-value, namely

$$P(F_{(p,n-p-1)} \geq f_{\text{obs}} \mid H_0),$$

which is the probability, under the null hypothesis, of observing a value of the test statistic at least as large as the observed one. We usually fix a significance level (α) that is appropriate for the problem under study, in relation to the probability of rejecting H_0 when it is true (Type I error), and then derive the $(1-\alpha)$-quantile of the F distribution with the corresponding degrees of freedom. More details are given in Cicchitelli et al. (2016).

5.2.2 Remarks on the chi-squared and F distributions

The chi-squared distribution with r degrees of freedom, denoted by χ^2_r, is defined as the distribution of the sum of the squares of r independent standard normal random variables:

$$\sum_{i=1}^{r} Z_i^2 \sim \chi^2_r$$

where $Z_i \sim \mathcal{N}(0,1)$ and the variables are mutually independent.

The probability density function of a chi-squared random variable with r degrees of freedom is shown in Figure 16.

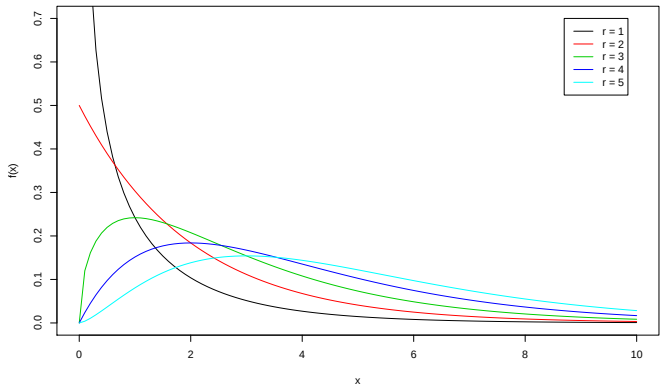


Figure 16: *Probability density function of a chi-squared random variable with different degrees of freedom*

For $Y \sim \chi^2_r$, we have $\mathbb{E}(Y) = r$ and $\mathbb{V}\mathrm{ar}(Y) = 2r$.

The Fisher–Snedecor distribution (named after Ronald Fisher and George W. Snedecor) is a continuous probability distribution. The probability density function of this random variable, with d_1 and d_2 degrees of freedom, is shown in Figure 17.

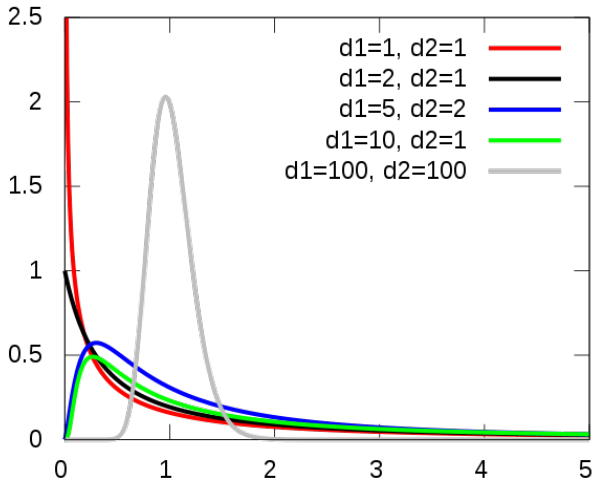


Figure 17: *Probability density function of an F random variable with different degrees of freedom*

5.2.3 Confidence intervals for the regression parameters

The least squares estimator is a linear combination of the response variables, and inference on the model parameters is conducted under the assumptions made on the error terms. As illustrated above, the least squares estimators are unbiased and have minimum variance. They follow a multivariate Gaussian distribution:

$$\hat{\boldsymbol{\beta}} \sim N_{p+1}(\boldsymbol{\beta}, \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}).$$

Each $\hat{\beta}_k$ has variance $\sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}$, where $(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}$ is the k-th diagonal element of the matrix $(\boldsymbol{X}'\boldsymbol{X})^{-1}$.

A confidence interval for β_k at confidence level $(1-\alpha)$ is

$$\hat{\beta}_k \pm t^*_{(1-\alpha/2,\, n-p-1)} \sqrt{S^2(\boldsymbol{X}'\boldsymbol{X})^{-1}_{kk}},$$

where $t^*_{(1-\alpha/2,\, n-p-1)}$ is the corresponding quantile of the Student's t distribution.

5.3 t-test for each regression parameter

We consider the significance test

$$H_0 : \beta_k = \beta_{H_0},$$

for each β_k, against

$$H_1 : \beta_k \neq \beta_{H_0}.$$

The test statistic is

$$t_{\beta_k} = \frac{\hat{\beta}_k - \beta_{H_0}}{SE_{\hat{\beta}_k}},$$

which represents the number of standard errors by which $\hat{\beta}_k$ differs from the value specified under H_0. In the special case in which $H_0 : \beta_k = \beta_{H_0} = 0$, it measures how many standard errors $\hat{\beta}_k$ is away from zero.

Under the null hypothesis, the test statistic follows a Student's t distribution with

$$df = n - p - 1.$$

Assuming that the null hypothesis H_0 is true, we compute the p-value

$$P(|T| \geq |t_{\text{obs}}| \mid H_0).$$

This is the probability of observing a value of the test statistic at least as extreme as the observed one, under the null hypothesis. It corresponds to the total area in the two tails of the reference distribution. Usually, software also reports significance codes associated with conventional thresholds, to summarize the strength of the evidence against the null hypothesis.

Strong evidence from the global F-test that at least one $\beta_k \neq 0$ does not necessarily imply that any individual t-test will identify a statistically significant effect.

Statistical software generally reports p-values together with significance codes. For example, Figure 18 shows the R Core Team (R Core Team, 2021) output obtained by fitting a multiple linear regression model with two explanatory variables.

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.82453    0.10657  -7.737 5.72e-14 ***
## X2           3.04361    0.04506  67.551  < 2e-16 ***
## X3          -0.59457    0.06229  -9.545  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

Figure 18: *Multiple regression results for data with two covariates (x_2 and x_3), obtained using the R function* `lm`.

Note that the significance codes shown in Figure 18 indicate conventional thresholds. A highly significant coefficient is commonly denoted by three stars (Grigoletto et al., 2017). A dot indicates a p-value in the interval $[0.05, 0.10)$. If no symbol appears next to the p-value, then the p-value is greater than 0.1 and the evidence against the null hypothesis is weak.

A remark on hypothesis testing

To obtain a probabilistic summary of the *evidence against* H_0, we consider the sampling distribution of the test statistic under the assumption that H_0 is true. We then evaluate how far into the tail of this distribution the observed test statistic falls. This is summarized by the tail probability of observing that value, or a more extreme one, in the direction specified by H_1 (the alternative hypothesis). This probability is called the p-value.

The p-value is the probability, assuming that H_0 is true, of observing a test statistic equal to the observed value or more extreme in the direction specified by H_1. The smaller the p-value, the stronger the evidence that the data provide against H_0 and in favor of H_1.

A small p-value, such as 0.01, means that the observed data would be unusual if H_0 were true. A moderate or large p-value, such as 0.26 or 0.83, means that the data are consistent with H_0; if H_0 were true, the observed data would not be unusual. *Smaller p-values reflect stronger evidence against H_0.*

Furthermore, in multiple linear regression, the value of each coefficient depends on the other covariates included in the model to explain the variability of the response. When these covariates are highly correlated, the results of this univariate statistical test may differ from those obtained after fitting a model with fewer covariates.

5.4 Multicollinearity: Nearly redundant explanatory variables

Collinearity occurs when explanatory variables are strongly linearly linked to each other. This creates an additional difficulty in separating the effect of each explanatory variable on the response. In order to provide reasonable estimates of the model parameters, $\boldsymbol{X}'\boldsymbol{X}$ must be invertible, which requires the covariate matrix to have full column rank. In other words, the columns of $\boldsymbol{X}$ must be linearly independent. Perfect multicollinearity makes the $\boldsymbol{X}'\boldsymbol{X}$ matrix non-invertible and implies that there are no unique least-squares estimates. Figure 19 illustrates the problem with near-perfect collinearity in the case of two explanatory variables, for which many planes yield the same minimum sum of

squared residuals.

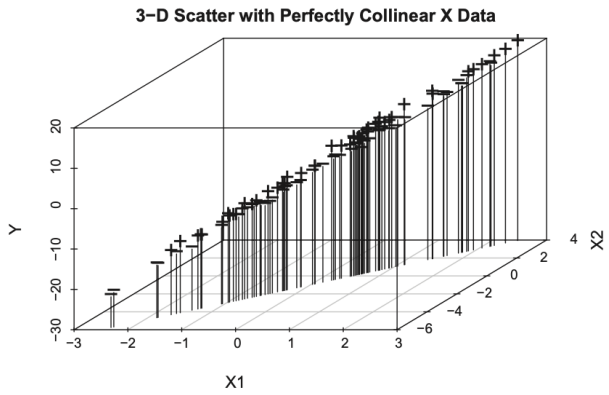


Figure 19: *3D Scatterplot of data where the two covariates are perfectly collinear (From Figure 8.6 of Westfall and Arias (2020)*

In this context, even when the matrix is invertible, there may still be insufficient variability in the data; that is, the columns may be nearly linearly dependent. This implies that the standard errors are relatively large (i.e., inflated) and that statistics such as the t-test become unreliable. In addition, a small change in the values of the covariates included in the model may lead to a substantial change in the estimated regression coefficients, and their signs may even be opposite to what intuition about the effect of the predictors would suggest.

Collinearity can be detected in several ways. Examining the correlation matrix of the predictors may reveal large pairwise correlations among them.

5.4.1 Variance Inflation Factor

One measure to assess whether a given explanatory variable can be well predicted by the other covariates is the Variance Inflation Factor (VIF, Faraway (2016)) expressed as

$$VIF(\hat{\beta}_k) = \frac{1}{1 - R_k^2},$$

where R_k^2 denotes the R^2 obtained by regressing x_k on the other explanatory variables in the model. The VIF represents the multiplicative increase in the variance of the variance

of $\hat{\beta}_k$ due to the linear dependence of x_k on the other covariates. In other words, the VIF measures how much the variance of $\hat{\beta}_k$ is inflated because of multicollinearity.

If R_k is close to one, the VIF will be large, indicating the presence of multicollinearity. In general, a high VIF value (typically, a VIF greater than 10 is considered high) indicates that the explanatory variables are strongly linearly associated. Once the VIF is particularly high, it maybe necessary to:

(i) construct a single combined variable from the highly correlated variables, for example by creating a new variable derived from the original ones; alternatively, one may increase the number of available observations and check whether the problem persists;

(ii) omit some variables, since the information they provide is redundant;

(iii) use an alternative approach to estimate the model parameters, such as ridge regression.

6 Variable selection: Data-driven procedures

In many applications, a large, potentially enormous, number of candidate predictor variables are available, and we must decide which of these variables should be included in the regression model. In the following, we consider the information criteria that help us choose the model that provides the best fit while remaining as parsimonious as possible in terms of the number of parameters. The aim is to construct a model that either predicts well or explains the associations in the data.

A model with many explanatory variables, and therefore many parameters, generally reduces the residual variance. Information criteria help select the most useful covariates from among those available by jointly considering both the fit and parsimony. We seek to minimize the criterion, since it balances model fit against model complexity. Note that if there are p potential predictors or covariates, there are 2^p possible models. These criteria help identify the best model among them.

Model selection is important for several reasons:

- First, omitting relevant variables can lead to biased coefficient estimates, especially when the omitted variables are correlated with the included predictors. By contrast, including irrelevant variables does not necessarily introduce bias, but it does increase the variance of the estimators and therefore reduces their precision. Variable selection is particularly important in observational studies because of confounding.

- Second, model specification and the choice of variables included in the model have a direct impact on predictive performance. The bias–variance trade-off in prediction is not unique to linear models, but is a general feature of statistical modelling. Variable selection is therefore crucial not only for prediction accuracy, but also for the applicability and interpretability of prediction models.

- Among the data-driven approaches to variable selection, one of the most widely used is stepwise selection. This procedure is mainly based on information-criterion methods derived from information theory, where entropy is used as a measure of discrepancy between two distributions (see the Kullback-Liebler distance, Kullback, 1959)). The most commonly used criteria are the (Bayesian Information Criterion, BIC, Schwarz, 1978), proposed by Schwarz, and the (Akaike Information Criterion, AIC, Akaike, 1973).

The BIC index, written in terms of the log-likelihood function, is defined as:

$$BIC = -2(loglikelihood) + \log(n) \times (\#par),$$

or

$$BIC = -2\hat{\ell}(\boldsymbol{\theta}) + (\#par)\log(n).$$

In other words, BIC is equal to minus twice the maximized log-likelihood, plus a penalty term that depends on the number of model parameters ($\#par$) and the logarithm of the sample size (n). Here, the log-likelihood is evaluated at the maximum likelihood

estimate of the parameter vector. The penalty increases with the number of parameters in the model, thereby discouraging unnecessarily complex specifications.

Because a higher likelihood indicates a better fit, smaller values of BIC correspond to better models once model complexity has been taken into account. If adding an extra parameter leads to only a slight improvement in the likelihood, the penalty term may outweigh that gain, resulting in a larger BIC value. For this reason, although the model including all available variables typically provides the best in-sample fit and the highest R-squared, BIC often favours a more reduced model, that is, a model including only a subset of the original variables and therefore fewer parameters.

Akaike's information criterion written with respect to model deviance is instead defined as follows:

$$AIC = -2(loglikelihood) + 2 \times (\#par),$$

which can be also expressed as:

$$AIC = n \log(\boldsymbol{z}'\boldsymbol{z}) + 2\#par,$$

where $Dev(\boldsymbol{z})$ is the residual deviance of the model. Again, the best model is the one with the lowest (higher) value of this index.

Stepwise methods perform a restricted search through the space of potential models, using information criteria to choose among them. These criteria are evaluated for each model by adding or removing one covariate at a time and comparing successive models. The three most common procedures, which are mainly automatic search strategies, are *backward*, *forward*, and *stepwise* selection. *Backward* elimination is the simplest procedure, since it starts with all covariates included in the model and removes one eligible variable at each iteration if its contributionto the AIC is negligible. The procedure stops when all variables have been evaluated, yielding a more parsimonious model with fewer parameters (James et al., 2013). Backward elimination can identify variables that are predictive when considered jointly, even if they are not predictive when considered individually.

The *forward* procedure reverses the backward method: it starts with no covariates in the model and, for each predictor not yet included, evaluates the AIC obtained by adding that predictor to the model. The procedure continues until no additional covariate can be added.

The *stepwise* procedure combines the previous two approaches (Westfall and Arias, 2020). It addresses situations in which variables that were added or removed at an earlier stage may need to be reconsidered. There are several variations in how this procedure is implemented. Since information criteria may lead to the selection of models with different numbers and types of variables, substantive and application-specific considerations should also guide the final choice of model.

We briefly mention other methods that can be used for model selection: (i) cross-validation, which mimics the partition of the data into a training set for parameter estimation and a validation set for assessing predictive performance; and (ii) Mallow C_p which introduce another form of penalization for including additional covariates in the model.

It is important to remember that none of these methods is designed to identify variables that are *causally related to the response*. It is also essential not to confuse *practical importance* of *statistical significance*, typically assessed through the univariate t-test on individual regression coefficients, with the importance of a covariate as evaluated by information criteria. Discarding covariates solely on the basis of t-test esults *promotes fundamental misunderstanding of statistics, since it suggests that an "insignificant" result implies "no effect"*.

7 Prediction

In many applications, the goal is to tailor a model to describe the observed data and predict the response for a new observation drawn from the same population, outside the sample. Machine learning methods are often used for this purpose, typically sacrificing interpretability in exchange for high predictive accuracy.

The model is used to estimate $\mathbb{E}(\boldsymbol{Y}|\boldsymbol{X} = \boldsymbol{x})$ representing the conditional expected value of the response given set of covariates. In addition to a point prediction, it is also essential to quantify the uncertainty associated with that prediction. A model should therefore be evaluated not only in terms of the accuracy of its predictions, but also in terms of the reliability of the corresponding uncertainty bounds.

For example, consider a model designed to predict the high-water mark of a river in order to build protective barriers. These barriers must be high enough to withstand floods that may substantially exceed the predicted maximum level.

Within the framework of the multiple linear regression model, predictive values can be derived in two different settings:

(i) for a unit that was not included in the original sample used to estimate the model parameters, that is, the value of the response variable for a new unit (prediction of a future value);

(ii) for the average value of the response (prediction of the mean response).

It should be noted that, if the estimated model is inadequate or it its assumptions are not satisfied, the resulting predictions will also be unreliable. In case (i), the aim it to predict the response value Y_c for a new statistical unit c on the basis of the estimated model; this is referred to as *the prediction of a future value*. For example, suppose we use a linear regression model for predicting house prices in a given area based on the number of bedrooms and proximity to the university. If a new house c enters the market with a given number of bedrooms x_{1c} and a given distance x_{2c} from the university, its selling price can be predicted under the assumption that the random response variable Y_c for that house satisfies the regression model. That is

$$\begin{aligned} Y_c &= \beta_0 + \beta_1 x_{1c} + \beta_2 x_{2c} + \epsilon_c \\ &= \mu_{y_c} + \epsilon_c, \end{aligned}$$

where ϵ_c is the error term associated with a new observation and $\epsilon_c \sim N(0, \sigma^2)$. The best

prediction $\tilde{Y}_c$ is obtained by minimizing the following prediction error

$$\tilde{\epsilon_c} = (Y_c - \tilde{Y}_c).$$

The result is $\tilde{Y}_c = \hat{\mu}_{y_c}$ obtained using the estimated regression coefficients and assuming $\mathbb{E}(\tilde{\epsilon_c}) = 0$

$$\hat{\mu}_{y_c} = \hat{\beta}_0 + \hat{\beta}_1 x_{1c} + \hat{\beta}_2 x_{2c}.$$

The expected value of the prediction error is

$$\mathbb{E}[(\tilde{Y}_c - Y_c)^2],$$

and the variance is

$$\mathbb{E}[(\tilde{Y}_c - Y_c)^2] = \text{var}(\tilde{Y}_c) + \text{var}(Y_c),$$

where $\text{var}(\tilde{Y}_c)$ denotes the variance of $\hat{\mu}_{y_c}$ and depends on the variability of the estimated regression coefficients, while $\text{var}(Y_c)$ denotes the variance of the new prediction, which, under the model assumptions, is equal to σ^2. Therefore, the overall variability combines the inherent variability of an individual observation, which cannot be reduced (σ^2, also called *the irreducible prediction error*), with the variability due to the uncertainty arising from estimating μ by $\hat{\mu}$, which can be reduced if the model provides a better fit. In matrix notation we have

$$\mathbb{E}[(\tilde{Y}_c - Y_c)^2] = \sigma^2 \boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c + \sigma^2,$$

where σ^2 is unknown and must be estimated, the above quantity is referred to as the standard forecast error when σ^2 is replaced by S^2. Hence, σ^2 also serves as a measure of prediction accuracy (McCullagh, 2002). More generally, a prediction interval is constructing for the random variable $\tilde{Y}_c$ with bounds that have a high probability of defining plausible values of the response variable for the new unit arising from the same population model.

(case i) The following pivotal quantity has a Student's t distribution with $n-p-1$ degrees of freedom and is used to construct a prediction interval for a future response:

$$\frac{Y_c - \tilde{Y}_c}{S\sqrt{\mathbf{1} + \boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c}} \sim t_{(n-p-1)}.$$

A $100(1-\alpha)\%$ prediction interval for a future response is therefore given by the following bounds:

$$\left[\tilde{Y}_c - t_{n-p-1;(1-\alpha/2)} S\sqrt{\mathbf{1} + \boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c},\ \tilde{Y}_c + t_{n-p-1;(1-\alpha/2)} S\sqrt{\mathbf{1} + \boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c}\right].$$

(case ii) The prediction interval for the mean response addresses the following question: "What would a house with characteristics x_c sell for, on average?". In this case, the estimate is more accurate because one source of variability is removed. Indeed, only the variance of $\hat{\boldsymbol{\beta}}$ due to the estimation of the regression coefficients has to be considered, whereas the variability associated with the individual random errors averages out of zero.

A $100(1-\alpha)\%$ confidence interval for the mean response is:

$$\left[\tilde{Y}_c - t_{n-p-1;(1-\alpha/2)} S\sqrt{\boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c},\ \tilde{Y}_c + t_{n-p-1;(1-\alpha/2)} S\sqrt{\boldsymbol{x}_c'(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{x}_c}\right].$$

Even though the two intervals may at first appear similar, they are in fact conceptually very different. In the second case, we constructed a confidence interval for the mean response; this means that the random interval contains the true mean value with probability $(1-\alpha)$. In the first case, by construct, we constructed an interval that is likely, more precisely with probability $1-\alpha$, to contain a future random observation.

7.1 Bootstrapping

We note that, when the model assumptions are not satisfied, one robust alternative for obtaining standard errors is bootstrapping. As illustrated in Section 3.4, the observed sample data can be used to estimate the standard errors for the parameters through the following steps:

(i) resample the units with replacement and then select the corresponding observations from the sample;

(ii) estimate the multiple linear regression model using only the bootstrap sample;

(iii) repeat the previous steps many times;

(iv) plot the histogram of the bootstrap distribution obtained for each parameter, such as β_0, β_1 and so on;

(v) compute the bootstrap confidence interval using the percentile method.

There are many variations of the bootstrap procedure, such as resampling residuals instead of observations, but these methods are beyond the scope of the present notes.

8 Strategies for model building

Multiple linear regression can be useful for the following tasks:

- explain the variability of the response variable in terms of the covariates (or explanatory variables), and estimating their effects on the response; that is, quantifying the expected change in the response variable associated with a one unit change in an explanatory variable, holding the other variables in the model constant;

- identifying the covariates that are the most relevant for understanding changes in the response;

- predicting the value of the response for new (out-of-sample) units, or estimating the mean response for given values of the explanatory variables.

In constructing the model, the following sequential steps should be considered, (see among others Newbold et al., 2013), guided by the underlying statistical theory:

1) **Model specification**: This step includes describing the problem under study, analysing the observed data by means of descriptive statistics, and selecting the response variable and covariates, first on theoretical grounds and then in light of the available data. It also requires a careful review of previous studies on the subject and consultation with experts in the relevant field of application. Appropriate assumptions under the postulated linear regression model must be also formulated.

2) **Model estimation**: This step involves estimating the model parameters, including the regression coefficients. If the model is intended primarily for prediction, a perfect fit is not expected; rather, the aim is to reduce prediction error. Information criteria, which jointly consider goodness of fit and adequacy and parsimony, can help in selecting a more suitable model according to the purpose of the analysis.

3) **Diagnostic checking**: This is mainly carried out by examining the regression residuals to order to assess possible violations of the basic assumptions of the model. For example, the model may be extended by including additional parameters, such as interaction effects or squared terms of the explanatory variables, or variables may be transformed by considering deviations from the mean or the logarithms. In such cases, one may return to step 1, or proceed to the next step. If the model shows poor adequacy, alternative regression models may be considered, such as generalized additive models for location, scale, and shape (GAMLSS).

4) **Interpretation and inference**: once the model has been validated and can be considered adequate for representing the phenomenon under study, it is necessary to report and interpret the point estimates and interval estimates of the regression coefficients, determine the values predicted by the fitted model, and summarise results appropriately.

5) **Archiving the data and code**: This is a fundamental step for ensuring the reproducibility and replicability of the results. This is especially important in the context of scientific research[8].

Note that statistical models are essential in data analysis because they do not only provide predictions, but also help us understand the associations among variables, assess the significance of the effects, and quantify uncertainty. By contrast, automatic machine learning methods often focus mainly on predictive performance, with limited interpretability and inferential insight.

[8]For more details see `https://assets.pubpub.org/vikcmjs4/ff3ab400-80d5-494a-a997-adbd0c486447.pdf`.

9 Categorical explanatory variables

Categorical covariates, also called factors, are often included in regression models. Suppose we are interested in the effect of firm size on the level of profits. The number of employees can be grouped into categories, thus defining a categorical variable for firm size with three levels ($J = 3$ categories): small, medium, and large. Another variable of interest may be binary, indicating whether or not the firm is listed on the stock exchange.

Analysis of covariance (ANCOVA) refers to multiple linear regression model that includes both qualitative and quantitative covariates. Categorical variables, such as binary variables, must be coded according to their levels. For example, in dummy coding, $j = 0$ may indicate the absence of a given characteristic and $j = 1$ its presence, although other coding schemes may also be used.

The estimation, inferential, and diagnostic procedures are the same as in the previous model, but special care is required in interpreting the parameters associated with categorical covariates. To ensure identifiability, one category is selected as a baseline category or reference category, and its parameter is set to 0. In R, the first category is used as the reference by default. The intercept then, represents the expected response for the reference category, while the other coefficients represents differences between each of the remaining categories and the reference category. For example, with $J = 4$ categories the model is written as

$$Y_i = \beta_0 + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 x_{i4} + \epsilon_i \qquad i = 1, \ldots, n,$$

where x_i is a dummy variable with $x_{ij} = 1$ equal to 1 if the ith unit belongs to category j, and 0 otherwise. These binary variables represent the categories of the qualitative covariate. As can be seen from the model specification, one category, in this case $j = 1$, is omitted, and the regression coefficients are estimated under the constraint that the corresponding parameter is equal to $\beta_1 = 0$. In this setting, the intercept β_0 represents the expected value of the response for the reference category, that is, $\mathbb{E}[Y|x_1 = 1, x_2 = 0, x_3 = 0, x_4 = 0] = \beta_0$ for units belonging to the first level. Differences

between groups ($\beta_j - \beta_k$), can then be expressed in terms of the regression coefficients. For example, β_2 represents the difference between the mean response for the second level and that for the first level, β_3 the difference between the third level and the first, and β_4 the difference between the fourth level and the first.

When the categorical covariate is binary, as in the example of firms where profits (Y) are explained by revenues (x_1) and by whether the firm is listed on the stock exchange (x_2) the observations are effectively divided into two groups. For firms that are not listed on the stock exchange ($x_2 = 0$), the expected value of the response is described by a regression line with intercept β_0 and slope β_1 as

$$Y_i = \beta_0 + \beta_1 x_{i1} + \delta_2 x_{i2} + \epsilon_i,$$

$$\mathbb{E}[Y|x_1, x_2 = 0] = \beta_0 + \beta_1 x_{i1}.$$

For firms that are listed ($x_2 = 1$), the expected value is described by a regression line with the same slope β_1 but with intercept $beta_0 + delta_2$ that is:

$$\mathbb{E}[Y|x_1, x_2 = 1] = \beta_0 + \beta_1 x_{i1} + \delta_2 = (\beta_0 + \delta_2) + \beta_1 x_{i1}.$$

Hence, the difference between listed and unlisted firms is constant and equal to δ_2. The model is therefore referred to as parallel-lines model. The partial regression coefficient β_1 measures the linear association between revenues and profits, independently of listing status, while δ_2 captures the difference in expected profits between listed and unlisted firms. This allows us to test whether listing on the stock exchange is associated with a significant difference in profits.

In summary, in this context:

- one category must be omitted from the model in order to ensure identifiability;
- the mean of the response variable is interpreted with reference to the omitted category;
- the estimated coefficients associated with the categorical variable are interpreted

as differences relative to the omitted category;

- the fitted values, R-squared, and the other model results remain the same regardless of which category is chosen as the reference.

10 Extension of the classical linear model

The linear model introduced above can be extended in several directions. *Generalized linear models* (McCullagh, 2019; McCullagh and Nelder, 1989) are commonly used when some assumptions of the classical linear model, such as normality. In the model specification presented earlier, we assumed that $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma \boldsymbol{I})$, implying that the errors are uncorrelated and homoscedastic. More general linear models allow for correlated and heteroscedastic error terms. We refer to Fahrmeir et al. (2022) for a comprehensive overview of generalized linear models.

Other extensions include generalized additive models (Hastie and Tibshirani, 1986) which allow for nonlinear smooth functions of the explanatory variables. Moreover, mixed effects models, make it is possible to account for potential correlations between observations through random intercepts and random slopes (Laird and Ware, 1982). More recently, generalized additive models for location, scale, and shape allow both the mean and the variance of the model to depend on the explanatory variables (Stasinopoulos et al., 2024).

In the following, we introduce one of the most widely used generalized linear models, namely the logistic regression model, which assumes a binomial distribution a binary response variable. First, we recall Bernoulli and the binomial distributions, and then we present the model formulation.

11 Recall of Bernoulli and Binomial distributions

11.1 Bernoulli distribution

A random experiment with only two possible outcomes, usually called success and failure, is known as Bernoulli trial. For example, when flipping a coin, the two possible outcomes are heads and tails. As another example, let X denote the daily average concentration of the atmospheric particles PM_{10}, and consider the event $X \leq 5$ (μg/m3), which represent the alert threshold, versus the event or $X > 5$. In such cases, the sample space of a Bernoulli trial can be denoted by $\Omega = \{s, f\}$ and we also assume that

$$P(s) = p, \quad P(f) = 1 - p = q.$$

The random variable X that maps outcome s to 1, and outcome f to 0 such that

$$X(s) = 1, \quad X(f) = 0$$

is called a Bernoulli random variable that is $X \sim$ Bernoulli(p), where p is the probability of success. The mean and variance of X are $\mathbb{E}(X) = p$, and $\mathbb{V}(X) = pq$.

11.2 Binomial distribution

In the case of repeated Bernoulli trials run independently n times with success probability p we consider X be the number of successes with support $\{0, 1, 2, ..., n\}$. X is called a binomial random variable and n and p are the parameters of X: n is called the size parameter, and p is the probability of success. We usually write:

$$X \sim \text{Binomial}(n, p) \quad \text{or} \quad X \sim B(n, p).$$

If $X \sim B(n,p)$, then its support is $\{0,1,2,...,n\}$ and its probability mass function is

$$f(k) = P(X=k) \quad = \quad \begin{cases} \frac{n!}{k!(n-k)!}\, p^k q^{n-k} & k \in \{0,1,2,...,n\} \\ 0 & \text{otherwise.} \end{cases}$$

This distribution takes its name from Newton's binomial theorem. Let $X = \sum_{i=1}^{n} Y_i$, where each Y_i is a Bernoulli random variable. Then, X represents the number of successes in n independent trials, each with the same probability of success. It follows that the probability of observing k success is

$$P(X=k) = \binom{n}{k} p^n (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^n (1-p)^{n-k}$$

where $k = 0, 1, \ldots, n$. The mean and the variance are given by $\mathbb{E}(X) = np$ and $\mathbb{V}(X) = npq$. Figure 20 shows the probability mass functions of the Binomial distribution for an increasing number of trials $n = 5, 10, 50$, with the probability of success equal to 0.5. The binomial distribution is symmetric when $p = 0.5$, but it becomes increasingly skewed as p approaches 0 or 1, especially when n is small.

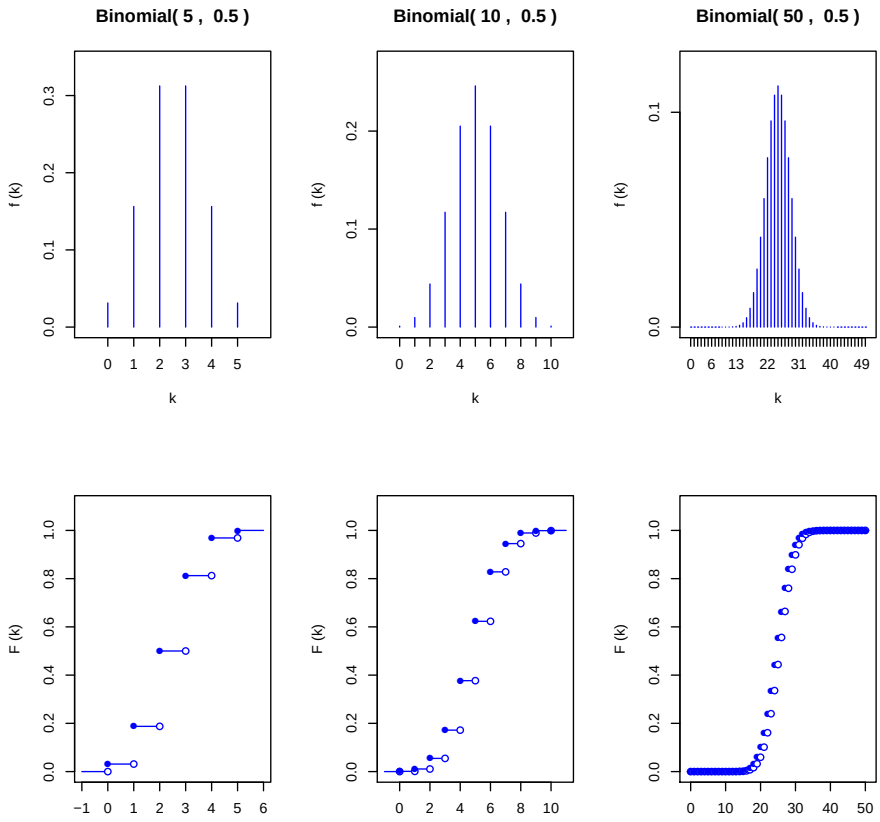


Figure 20: *Probability mass function and cumulative distribution function of a binomial distribution for an increasing number of trials* $n = 5, 10, 50$*, with the probability of success equal to 0.5.*

12 Logistic regression for binary data

Logistic regression is a model for a binary dependent variable explained by p explanatory variables, which may be either quantitative or qualitative. The sample observations are assumed to be realisations from a binomial distribution. This model is widely used in many fields, such as marketing research, to model consumer decisions, for example the choice between two brands.

The logit of the probability of interest (success) is expressed as a linear function of the explanatory variables, which may be either continuous or categorical. Continuous variables can be included as deviations from their mean or through transformations such as squared terms. Binary or categorical variables may also be included, along with interaction terms.

In the following, let $Y_1, \ldots, Y_n$ denote the binary response variables, each associated with a probability p of belonging to the reference category (success =1). Specifically, for unit i let this probability be denoted by p_i

$$P(Y_i = 1) = p_i, \quad i = 1, \ldots, n.$$

Assume that Y follows a Bernoulli distribution, $Y \sim Bern(p)$, with probability mass function

$$P(Y = y) = p^y (1-p)^{1-y}, \quad y \in \{0, 1\}.$$

We consider the following measure named odds of success

$$O = odds = \left(\frac{P(Y_i = 1)}{P(Y_i = 0)} \right),$$

which is the ratio between the probability of success and failure

$$odds = \frac{p}{1-p}.$$

The odds are commonly used in betting games such as horse racing, where they repre-

sent the ratio of the probability of success to the probability of failure (for example odds of 3 to 1 indicate three successes for every failure). The odds are always non-negative and increase as the probability of success becomes larger relative to the probability of failure.

The natural logarithm of the odds is the logit

$$\log(O) = \log\left(\frac{p}{1-p}\right).$$

Another measure is the odds ratio or cross product ratio which is defined as

$$OR = \frac{O_2}{O_1} = \frac{p_2/(1-p_2)}{p_1/(1-p_1)},$$

and it ranges between $(0, +\infty)$. It is equal to 1, the odds are the same in the two groups, and therefore so are the corresponding probabilities. If $OR > 1$, the odds are greater in group 2 than in group 1; if $OR < 1$ the odds are smaller in group 2 than in group 1. If the odds ratio is equal to 1, the two underlying random variables are independent.

The logistic regression model for binary data is formulated by considering a linear model for the logit the probability of success: $p_i = P(Y_i = 1)$

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \boldsymbol{x}_i'\boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip},$$

where $\boldsymbol{x}_i$ denotes the column vector of p explanatory variables $(x_{i1}, \ldots, x_{ip})$ observed for unit $i, i = 1, \ldots, n$. Since the model is linear in the logit, its inverse, called the expit, represents the probability of success for the i-th unit

$$P(Y_i = 1|\boldsymbol{x}_i) = p_i = \frac{\exp(\boldsymbol{x}_i'\boldsymbol{\beta})}{1 + \exp(\boldsymbol{x}_i'\boldsymbol{\beta})}.$$

If the coefficients β are positive, the probability function increases, while always remaining between 0 and 1. Moreover, the larger the value of β, the steeper the increase. When all explanatory variables are equal to 0, the conditional expected value of the response is $\exp(\beta_0)/(1 + \exp(\beta_0)$ and if $\beta_0 = 0$ then the intercept is 0.5.

The model with two covariates is

$$\log \frac{P(Y_i = 1)}{P(Y_i = 0)} = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}.$$

Then the failure probability can be expressed as

$$P(Y_i = 0|\boldsymbol{x}_i) = 1 - P(Y_i = 1|\boldsymbol{x}_i) = \frac{1}{1 + \exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})}.$$

Note that the odds with this model are

$$\frac{P(Y_i = 1)}{P(Y_i = 0)} = \exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) = e^{(\beta_0)} e^{(\beta_1)^{x_1}} e^{(\beta_2)^{x_2}},$$

the odds have an exponential association with $\boldsymbol{x}$, since a one-unit increase in x has a multiplicative effect of e^{β}. Each e^{β_j} represents the conditional odds ratio, that is, the ratio of the odds at $x_j + 1$ to the odds at $x_j = u$, holding the other explanatory variables constant.

The regression coefficients can be interpreted as follows:

- $\exp(\beta_0)$ represents the odds in favour of success rather than failure when all explanatory variables in the model are equal to zero;

- $\exp(\beta_1)$ represents the multiplicative change in the odds of success associated with a one-unit increase in the covariate, holding the other covariates constant.

  More generally, $\exp(\beta_1)$ describes the change in the ratio between the probability of success and the probability of failure for a one-unit increase in the variable x_i, with all other explanatory variables held fixed. Thus, a one-unit increase in x produces a multiplicative change of $\exp(\beta_1)$ in the odds of success (or betting ratios).

12.1 Inference

Assuming independence between observations, and given a fixed set of covariates $\boldsymbol{x}$ the likelihood function is expressed as follows

$$L(\boldsymbol{\theta}) = p(y_1|\boldsymbol{x})p(y_2|\boldsymbol{x})\ldots p(y_n|\boldsymbol{x}),$$

that is

$$L(\boldsymbol{\theta}) = \prod_i p(\boldsymbol{y}_i|\boldsymbol{x}).$$

The log-likelihood is

$$\ell(\boldsymbol{\theta}) = \log p(y_1|\boldsymbol{x}) + \log p(y_2|\boldsymbol{x}) + \ldots + \log p(y_n|\boldsymbol{x}),$$

$$\ell(\boldsymbol{\theta}) = \sum_i \log p(\boldsymbol{y}_i|\boldsymbol{x}).$$

As introduced in Section 3.2, maximum likelihood estimators are obtained by numerically maximizing the log-likelihood function. Since the estimating equations are non-linear in the parameters, iterative methods are required to maximize the log-likelihood function. Two algorithms are commonly used: the Newton- Raphson algorithm and the Iterative Proportional Fitting Algorithm. The Newton-Raphson algorithm approximates the score function linearly at a given point and then iteratively searches for the maximum of the log-likelihood function. The goodness of fit of the model is assessed using information criteria, including Akaike's Information Criterion (AIC), as illustrated in the previous chapter.

Due to the properties of the maximum likelihood estimator, we use the following test statistic to test the null hypothesis that a single regression coefficient is equal to zero, that is, $H_0 : \beta_k = 0$. Under the null hypothesis, we have

$$\frac{\hat{\beta}_k - 0}{aSE_k} = Z_n \rightarrow_d N(0, 1),$$

where aSE denotes the standard error, computed based on the asymptotic distribution

of the maximum likelihood estimators.

Based on asymptotic results, confidence intervals for the regression parameters are constructed at a given nominal level $(1-\alpha) = 0.95$ as

$$P\left(\frac{|\hat{\beta}_k - \beta_k|}{aSE_k} \leq 1.96\right)$$

therefore we obtain $\hat{\beta}_k \pm 1.96 \times a\hat{S}E_k$.

A comparison between nested models can be carried out using the likelihood ratio test, which is based on the deviance. Let $\ell(\hat{\boldsymbol{\theta}})$ denote the maximized log-likelihood of the fitted model, and let $\ell(\boldsymbol{y})$ denote the maximum achievable log-likelihood, that is, the value obtained under a *saturated model*, namely a model with the minimum possible degrees of freedom. The drawback of the saturated model is that it does not smooth the data and it is not parsimonious. It is used only as a benchmark for constructing a measure, called the likelihood ratio statistic, that compare the fitted model with the saturated one. The scaled deviance is defined as

$$D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}) = 2 \log \left[\frac{maximum\ likelihood\ for\ saturated\ model}{maximum\ likelihood\ for\ chosen\ model}\right]$$

$$D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}) = 2[\ell(\boldsymbol{y}) - \ell(\hat{\boldsymbol{\theta}})]$$

and it follows that $D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}) \geq 0$, since the saturated model is more general than the fitted model. The larger the value of $D(\boldsymbol{y}; \hat{\boldsymbol{\theta}})$, the poorer the fit of the model, see Agresti and Kateri (2021) for more details.

The scaled deviance can be used also to compare two nested models. Let M_0 denote a special case of M_1, where M_0 includes p_0 covariates and M_1 includes p_1 covariates. Since simpler models usually provide a poorer fit, we expect them to have larger deviances, that is $D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}_0) \geq D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}_1)$. We then use the following likelihood ratio test, under the null hypothesis that model M_0 holds, against the alternative hypothesis that the true

association is described by model M_1. The test statistic is

$$2[\ell(\hat{\boldsymbol{\theta}}_1) - \ell(\hat{\boldsymbol{\theta}}_0)] = D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}_0) - D(\boldsymbol{y}; \hat{\boldsymbol{\theta}}_1)$$

and it becomes larger when M_0 provides a poorer fit than M_1. In this case, differences between deviance statistics often have an asymptotic χ^2 null distributions (that is, when M_0 holds), with degrees of freedom equal to $p_1 - p_0$.

13 Model-based clustering and classification

Model-based clustering relies on the notion of finite mixture models, which are designed to model unobserved heterogeneity in the population. The underlying idea is to specify a joint model for the data, in which each group is represented by a particular mixture component, since each cluster is assumed to correspond to a specific subpopulation.

Model-based techniques for clustering and classification in statistics and machine learning are essential for exploiting the complex and massive amounts of data available today, especially newer data types such as high-dimensional, network, textual, and image data. With the rapid growth of data, clustering and classification tasks require automated algorithms for detecting cluster structures. In the 1960s, cluster analysis was placed on a statistical basis by framing clustering as inference for a finite mixture model. Most early clustering methods were based on measures of similarity between units. Nowadays, new types of data require classification into groups based on specific characteristics. These include genetic microarray data, retail barcode data, website-usage data, and image-analysis applications, such as detecting tumours in digital medical images.

It is important to emphasize the distinction between clustering and classification. Clustering is an unsupervised method that aims to divide a set of units into groups, whereas classification involves assigning units to pre-defined classes. This is typically done by assigning an observation to the class whose units it most closely resembles. The first statistical method for classification, due to Ronald Fisher in his famous work on discriminant analysis, is a classic example of this approach.

Finite mixture models are among the most widely used statistical approaches for clustering, since they can account for complex features such as multimodality, skewness, kurtosis, and unobserved heterogeneity. The oldest method, known as the latent class model, was proposed in sociology by Lazarsfeld in 1950 and developed for discrete multivariate data. For continuous data, a mixture of multivariate normal distributions was proposed by Wolfe in 1963; its parameters were estimated by maximum likelihood using an algorithm similar to the expectation-maximization algorithm. These approaches are

based on a probability model and therefore, compared with other clustering methods, they allow the model to be checked, the number of clusters to be chosen, and uncertainty about clustering and classification to be quantified.

13.1 Finite mixture models

Finite mixture distributions are applied to data with two main purposes in mind: to provide an appealing semi-parametric framework for modelling unknown distributional shapes, and to provide a probabilistic clustering of the data into a finite number of clusters.

Finite mixture models allow us to explore the structure of the data in inferential terms and to identify groups of units (clusters), by assuming the reference population is heterogeneous. They make it possible to classify the units into distinct groups that are homogeneous with respect to the characteristics of interest. This is similar to cluster analysis, which groups a set of units in such a way that the units in the same group, or cluster, are more similar to each other than to units in other groups.

Consider, for example, in a drug treatment in which patients may have very different reactions to the same drug, as it was in the case of adverse effects from COVID-19 vaccines. Asthma is treated with theophylline, and the dose depends on individual body weight. Therefore, a pharmacokinetic model, that is, a studying drug absorption and disposition for the development of new drugs, that assumes identical parameters for all subjects would be unrealistic.

These models are suitable for dealing with unobserved heterogeneity, which occurs when a sample is drawn from a statistical population without knowledge of the presence of underlying subpopulations. In this case, the mixture components can be interpreted as the densities of the subpopulations, while the mixing weights represent the proportions of each subpopulation in the overall population. As an example consider the fish length measurements, in inches, for snappers studies by Cassie in 1954. The data show a certain degree of heterogeneity, with the presence of several modes. A possible explanation is that the fish belong to different age groups, but age is difficult

to measure, so no information is collected on this characteristic. Mixtures of Gaussian distributions are a suitable model for analyzing these data.

In the following, we consider only mixture models with Gaussian components. These models are widely used because they are mathematically tractable, and maximum likelihood estimates of the parameters can be derived relatively easily. However, as discussed below, model selection, specifically, choosing the number of components (clusters), requires particular care.

13.2 Mixtures of Gaussian distributions

Mixtures of Gaussian distributions (McLachlan and Peel, 2000) are the most widely used models for continuous data and popular in statistical learning, pattern recognition, and data mining.

Let $\mathcal{D} = \{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$ be a n dimensional random vector where $\mathbf{x}_i \in \mathcal{X}$ are assumed to be independent and identically distributed with density function

$$f_\eta(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k f_{\theta_k}(\boldsymbol{x}),$$

for $k \in I\!N_k : \ f_{\theta_k} \in \{f_\theta : \boldsymbol{\theta} \in \Theta\}$ some parametric family which in this context will be the multivariate Gaussian density function with a mean vector $\boldsymbol{\mu}_k$ and a variance-covariance matrix, where:

- K is the number of mixture components,
- f_{θ_k} is the density of the kth component of the mixture,
- π_k are the mixture weights which are defined according to the following constraints $0 < \pi_k \leq 1, \ \sum_{k=1}^{K} \pi_k = 1$, $\boldsymbol{\theta} = (\pi_1, \ldots, \pi_K, \theta_1, \ldots, \theta_K)$.
- $\boldsymbol{\theta}$ represents the vector of parameters.

As previously stated clusters correspond to subpopulations distributed as f_{θ_k} with

different parameter values θ_k

$$f_\eta(\boldsymbol{x}) = \sum_{k=1}^{K} \pi_k f_{\theta_k}(\boldsymbol{x}).$$

Two-step version:

$$\begin{aligned} i \in I\!N_n : \; Z_i \quad \text{i.i.d.} \quad &\sim \quad \text{Multinomial}(1, \pi_1, \ldots, \pi_K), \\ \boldsymbol{x}_i | Z_i = k \quad &\sim \quad f_{\theta_{Z_i}}. \end{aligned}$$

With Bayes' formula, for classification:

$$P(Z_i = k | \boldsymbol{x}_i) = \frac{\pi_k f_{\theta_k}(\boldsymbol{x}_i)}{\sum_{j=1}^{K} \pi_j f_{\theta_j}(\boldsymbol{x}_i)}.$$

Once maximum likelihood estimation of the parameters has been obtained, a maximum a posteriori procedure (MAP) is applied assigning each $\boldsymbol{x}_j$ to the mixture component with the largest posterior conditional probability

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \prod_{i=1}^{n} \left(\sum_{k=1}^{K} \pi_k f_{\theta_k}(\boldsymbol{x}_i) \right).$$

Estimate posterior probability for labels:

$$\hat{p}_{ik} = \hat{P}(Z_i = k | \boldsymbol{x}_i) = \frac{\hat{\pi}_k f_{\hat{\theta}_k}(\boldsymbol{x}_i)}{\sum_{j=1}^{K} \hat{\pi}_j f_{\hat{\theta}_j}(\boldsymbol{x}_i)}.$$

Partitioning (if needed):

$$\hat{Z}_i = \arg\max_{k \in I\!N_k} \hat{p}_{ik}.$$

13.3 Parsimonious covariance decomposition

Gaussian populations are elliptical and can have flexible shapes. Data generated by a finite mixture model of Gaussian distributions are characterized by groups, or clusters,

centered at the mean $\boldsymbol{\mu}_k$, with higher density for points closer to the mean. However, within-cluster distances should be small. Geometric characteristics of the mixture

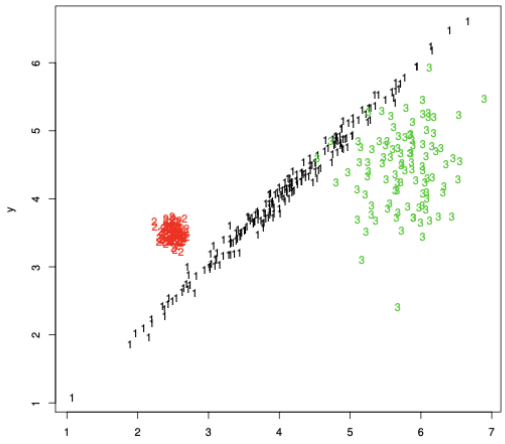


Figure 21: *Example of data with a mixture structure consisting of three components.*

components can be controlled by imposing constraints on the variance-covariance matrix through the eigen-decomposition, see Figure 22. The covariance matrix can be flexible Σ_k

$$\Sigma_j = \lambda_j D_j A_j D_j^T, \; j = 1, \ldots, K,$$

where

- $(\lambda_{j1}, \ldots, \lambda_{jp})$ eigenvalues
- $\lambda_j = \prod_{i=1}^{p} (\lambda_{ji})^{1/p}$ hypervolume, is a scalar controlling the volume
- D_j matrix of eigenvectors, is an orthogonal matrix of eigenvectors of $\boldsymbol{\Sigma}_k$ controlling the orientation
- $A_j = \frac{1}{\lambda_j} \text{diag}(\lambda_{j1}, \ldots, \lambda_{jp})$ is a diagonal matrix controlling the shape with $\det A_j = 1$.

Characteristics of the component distributions, such as *volume, shape, and orientation*, are usually estimated from the data and can be allowed to vary between clusters.

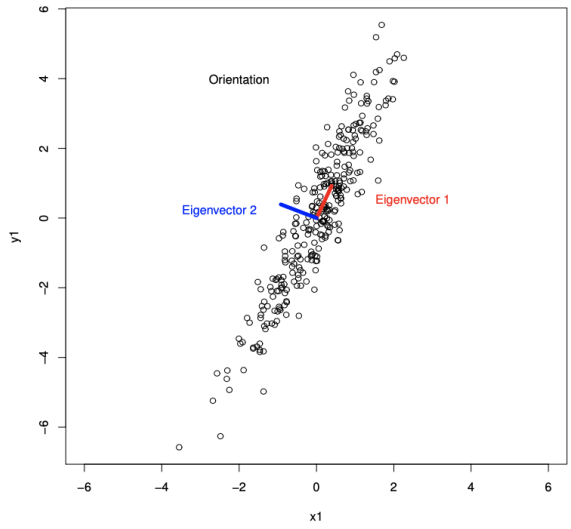


Figure 22: *Scatterplot with a over impose spectral decomposition.*

Using the coding provided within the R package mclust (Scrucca et al., 2023), we have "V" for variable, "E" for equal, "I" for the unit matrix. Models are defined by three letter codes corresponding to volume, shape, and orientation.

- For a *univariate mixture model*:
  E: equal variance (one-dimensional) and
  V: variable variance (one-dimensional)

- For the case of *multidimensional mixture model*, some possible specifications are the following:
  EII: spherical, equal volume; clusters are similar in terms of within-cluster dissimilarity/variation
  VII: spherical, unequal volume
  EEI: diagonal, equal volume and shape.

Figures 23, 24, and 25, illustrate the shapes derived from the densities of mixtures with these three different parameterizations of the variance-covariance matrix.

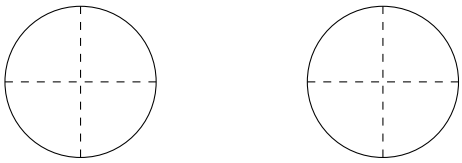


Figure 23: *Ellipses for two spherical models with equal volume, model label EII.*

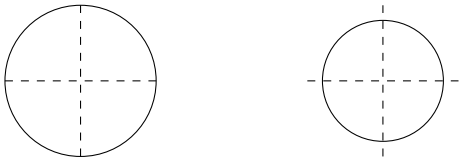


Figure 24: *Ellipses for two spherical models with unequal volume, model label VII.*



Figure 25: *Ellipses for two diagonal models with equal volume and shape, model label EEI.*

13.4 Maximum likelihood inference

Given a random sample of observations $\boldsymbol{x}_i, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n$, the log-likelihood of the finite mixture, assuming fixed K, is given by

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k f_{k,\Sigma_k}(\mathbf{x}_i) \right)$$

subject to $\pi_k > 0 \quad \forall k, \ \sum_{k=1}^{K} \pi_k = 1$, where $\boldsymbol{\theta}$ denotes the parameters to be estimated. There is no straightforward analytic solution, so an an algorithm is needed to find local optima.

The Expectation-Maximisation (EM) algorithm became popular in the 1970s through the work of Baum et al. (1970); Dempster et al. (1977), because it was formulated in terms of missing or incomplete data. However, it hav already been implicitly introduced McKendrick (1926) to estimate infection rates. The EM algorithm exploits the notion of incomplete or missing data and iteratively imputes them. It is used to estimate the parameters of the mixture model and with two iteratively repeated steps, the E-step and the M-step, allows the system of likelihood equations to be solved.

Some of the main advantages of the algorithm are the following:

- it is numerically stable,
- it often converges to finite local maxima even if the likelihood surface is unbounded
- it is quite simple to implement (E-step and M-step)
- both steps require little memory space.
- convergence of the algorithm can be monitored by checking the monotonicity of the increments of the likelihood function.

Convergence in the final steps is generally slow, although acceleration methods exist, and in some cases it is sensitive to the choice of initial parameter values. Once the

model parameters have been initialized, the algorithm alternates two steps: (i) an expectation step (E-step), in which the conditional expected value of the complete data log-likelihood is considered $\ell^*(\boldsymbol{\theta})$ is computed given the parameter values from the previous step and the observed data, and (ii) a maximization step (M-step), in which the parameters are updated by maximizing this expected value. To check the convergence, two common criteria are applied at each step: the relative difference between log-likelihood values of two consecutive iterations and the difference between the corresponding parameters vectors.

The choice of the number of mixture components and of the appropriate variance-covariance matrix can be addressed using the Bayesian Information Criterion (BIC). Another information criterion is the integrated completed-data likelihood (IC) (Biernacki et al., 2000), which penalizes the BIC through an entropy term measuring the overlap between clusters. It is formulated as

$$ICL = BIC_k + 2\sum_{i=1}^{n}\sum_{k=1}^{K} c_{ij}\log(\hat{z}_{ik})$$

where $\hat{z}_{ik}$ is the conditional probability that $\boldsymbol{x}_i$ arises from the kth mixture component and $c_{ij} = 1$ if the ith unit is assigned to cluster k, and zero otherwise.

Bootstrap methods can be applied to obtain standard errors for the parameter estimates. In this approach, several samples are repeatedly drawn from the fitted model, or from the observed data, and the mixture model is re-estimated for each bootstrap sample. The variability of the resulting estimates across bootstrap replications provides an empirical approximation of their sampling distribution, from which standard errors and confidence intervals can be obtained.

References

Agresti, A. and Kateri, M. (2021). *Foundations of Statistics for Data Scientists: With R and Python*. Chapman and Hall/CRC, Boca Raton.

Akaike, H. (1973). Information theory and an extension of the maximum likelihood principle. In Petrov, B. N. and Csaki, F., editors, *Second International symposium of information theory*, pages 267–281, Budapest. Akademiai Kiado.

Barnett, V. (1975). *Comparative Statistical Inference*. John Wiley and Sons.

Baum, L., Petrie, T., Soules, G., and Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *Annals of Mathematical Statistics*, 41:164–171.

Bentler, P. M. (1980). Multivariate analysis with latent variables: causal modeling. *Annual Review of Psychology*, 31:419–456.

Biernacki, C., Celeux, G., and Govaert, G. (2000). Assessing a mixture model for clustering with the integrated completed likelihood. *IEEE transactions on pattern analysis and machine intelligence*, 22:719–725.

Blitzstein, J. K. and Hwang, J. (2015). *Introduction to Probability*. CRC Press, Boca Raton, FL.

Cicchitelli, G., D'urso, P., and Minozzo, M. C. (2016). *Statistica: Principi e metodi*. Pearson.

Cox, D. (1995). Comment on: Discussion of the paper by Chatfield. *Journal of the Royal Statistical Association*, 158:455–465.

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34:187–202.

Davison, A. C. and Hinkley, D. V. (1997). *Bootstrap Methods and their Application*, volume 1. Cambridge University Press.

Dawid, A. P. (2002). Influence diagrams for causal modelling and inference. *International Statistical Review*, 70:161–189.

Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm (with discussion). *Journal of the Royal Statistical Society, Series B*, 39:1–38.

Efron, B. and Hastie, T. (2021). *Computer Age Statistical Inference, Student Edition: Algorithms, Evidence, and Data Dcience*, volume 6. Cambridge University Press.

Efron, B. and Tibshirani, R. J. (1994). *An introduction to the bootstrap*. CRC press.

Everitt, B. S. and Hand, D. J. (1981). *Finite Mixture Distributions*. Chapman and Hall, London.

Fahrmeir, L., Kneib, T., Lang, S., and Marx, B. D. (2022). *Regression: Models, methods and applications*. Springer.

Faraway, J. J. (2016). *Linear Models with R*. Chapman and Hall/CRC.

Gelman, A., Carlin, J., Stern, H.and Dunson, D., Vehtari, A., and Rubin, D. (2013). *Bayesian data analysis (3rd ed.)*. Chapman and Hall/CRC.

Grigoletto, M., Ventura, L., and Pauli, F. (2017). *Modello Lineare: Teoria e Applicazioni con R*. G Giappichelli Editore.

Hastie, T. and Tibshirani, R. (1986). Generalized Additive Models. *Statistical Science*, 32:297–3109.

Hastings, W. K. (1970). Monte Carlo sampling methods using Markov chains and their application. *Biometrika*, 57:97–109.

Hennig, C. (2010). Mathematical models and reality: A constructivist perspective. *Foundations of Science*, 15:29–48.

Hernán, M. and Robins, J. (2019). *Causal Inference*. Boca Raton: Chapman & Hall/CRC.

Holland, P. W. (1986). Statistics and causal inference. *Journal of the American Statistical Association*, 81:945–960.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). *An Introduction to Statistical Learning*. Springer.

Kass, R. E. (2011). Statistical inference: The big picture. *Statistical science: a review journal of the Institute of Mathematical Statistics*, 26:1.

Kullback, S. (1959). *Information Theory and Statistics*. J. Wiley and Sons, New York.

Laird, N. M. and Ware, J. H. (1982). Random-effects models for longitudinal data. *Biometrics*, 38:963–974.

Lazarsfeld, P. F. (1955). Interpretation of statistical relations as a research operation. In Lazarsfeld P. F., R. M., editor, *The Language of Social Research: a Reader in the Methodology of Social Research*, pages 115–125, Cambridge, MA. Glencoe, Free Press.

Markov, A. A. (1910). Investigation of the general case of trials associated into a chain [in russian]. *Zapiski. Akad. Nauk (St. Petersburg) Fiz.-Matem., (8th ser.*, 25:465–507.

McCullagh, P. (2002). What is a statistical model? *Annals of Statistics*, 30:1225–1310.

McCullagh, P. (2019). *Generalized Linear Models*. Routledge.

McCullagh, P. and Nelder, J. A. (1989). *Generalized Linear Models, 2nd Edition*. Chapman and Hall, CRC, London.

McKendrick, A. G. (1926). Application of mathematics to medical problems. In *Proceedings of the Edinburgh Mathematical Society*, pages 98–130.

McLachlan, G. and Peel, D. (2000). *Finite Mixture Models*. Wiley.

McNicholas, P. D. (2016). *Mixture Model-Based Classification*. Chapman and Hall, Boca Raton.

Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953). Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21:1087–1092.

Newbold, P., Carlson, W. L., and Thorne, B. (2013). *Statistics for Business and Economics*. Pearson Boston, MA.

Neyman, J. and Pearson, E. S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Phil. Trans. Roy. Soc. A*, 231:289–337.

Peng, R. D. and Parker, H. S. (2022). Perspective on data science. *Annual Review of Statistics and Its Application*, 9:1–20.

Quenouille, M. H. et al. (1949). Problems in plane sampling. *The Annals of Mathematical Statistics*, 20:355–375.

R Core Team (2021). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66:688.

Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6:461–464.

Scrucca, L., Fraley, C., Murphy, B., and Reftery, A. (2023). *Model-based, clustering, classification, and density estimation using mclust in R*. CRC Press, Boca Raton.

Stasinopoulos, M., Kneib, T., Klein, N., Mayr, A., and Heller, G. (2024). *Generalized Additive Models for Location, Scale and Shape: A Distributional Regression Approach, with Applications*. Cambridge University Press, New York.

Tukey, J. W. (1962). The future of data analysis. *The Annals of Mathematical Statistics*, 33:1–67.

Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley, Reading, MA.

Westfall, P. H. and Arias, A. L. (2020). *Understanding Regression Analysis: A Conditional Distribution Approach*. CRC Press.

Wing, J. M. (2020). Ten research challenge areas in data science. *arXiv preprint arXiv:2002.05658*.